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Ms. Katelene DA CRUZ DELGADO

Ex-post evaluation of fishery policy interventions on
Decapterus macarellus (Cuvier, 1833) stock recovery,
in northern Cabo Verde

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Jury Composition

Supervisor: Prof. KOUAMELAN Essetchi Paul, Dean of FBS-UFHB, Côte d'Ivoire

Co-Supervisor: Dr. OSEMWEGIE Isiememen, Senior Researcher at ZEF, UB, Germany

Tutor: Dr. MEDINA Aníbal, Senior Researcher at IMar, Cabo Verde

DEDICATION

K grand honra e ligria, um t dedica ess tese a nhes inspiraço de vida. P tud ques home d'boa fé e p ques amdjer d'luta k sempre inpiram, e t continua ta inpiram na nhe dia-a-dia, specialment p nhe pai e nha mãe, q desd ced insnam importancia de ser persistente e resiliente n vida!

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TABLE OF CONTENT

TABLE OF CONTENT	III
LIST OF ABBREVIATIONS AND ACRONYMS	IV
LIST OF FIGURES	V
LIST OF TABLES	VI
INTRODUCTION	1
CHAPTER 1 – BIBLIOGRAPHIC REVIEW	3
1.1 Description of the study area	3
1.2 Management fisheries in Cabo Verde.....	4
1.3 <i>Decapterus macarellus</i> (Cuvier, 1833).....	8
CHAPTER 2 - MATERIAL AND METHODS	10
2.1. Material.....	10
2.2. Field study methodology.....	10
2.2.1 Policy analysis method	10
2.2.2 Ecological methods for sampling and analyses	11
2.2.3 Social method to assess stakeholders' perception.....	13
CHAPTER 3 - RESULTS AND DISCUSSION.....	14
3.1 Fishery policies analysis	14
3.2 <i>Decapterus macarellus</i> ' stock state before and after the implementation of the fishing management policies	20
3.3 Stakeholders' perception about <i>D. macarellus</i> fishing management policies.....	26
CONCLUSION.....	33
RECOMMENDATIONS	35
PERPECTIVES.....	38
BIBLIOGRAPHIC REFERENCES.....	39
APPENDIX.....	47
Appendix I: Fish sampling sheet	47
Appendix II: Survey for fishery stakeholders.....	48
Appendix III: Stakeholders workshop	51
Appendix IV: Stakeholders workshop presence list	52
ANNEX.....	54
Annex I: Maturation Scale Guide for Pelagic Fish (Translated from IMar scale)	54

LIST OF ABBREVIATIONS AND ACRONYMS

Abbreviations	
APESC	Associação dos Armadores de Pesca (Fishing Shipowners Association)
APPECCAO	Projecto de Adaptação das Políticas de Pescas às Mudanças Climáticas em África Ocidental (Adapting Fisheries Policies to Climate Change in West Africa Project)
CG	Coastal Gard
EEZ	Economic Exclusive Zone
DNPA	Direção Nacional das Pescas e Aquacultura (National Directorate for Fisheries and Aquaculture)
EMAR	Escola do Mar (Sea School)
GDP	Gross Domestic Product
IGP	Inspeção Geral das Pescas (General Institute for Fisheries Inspection)
IKI	International Climate Initiative
IMAR	Instituto do Mar (Sea Institute of Research)
IMP	Instituto Marítimo Portuário (Maritime Port Institute)
INDP	Instituto Nacional das Pescas (former National Fisheries Institute)
IPBES	Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services
MM	Ministério do Mar (Sea Ministry)
MP	Marine Police
NGO	Non-Governmental organization
NP	National police
PANA II	Segundo Plano de Ação Nacional para o Ambiente (Second National Environmental Action Plan)
PEDS	Plano Executivo Para Desenvolvimento <i>Sustentável</i> /Sustentavel (Executive plan for Sustainable Development)
UTA	Universidade Técnica do Atlântico (Technique University of Atlantic)
WABES	Projecto de Adaptação das Políticas de Pescas às Mudanças Climáticas em África Ocidental (Adapting Fisheries Policies to Climate Change in West Africa Project)West Africa Biodiversity and Ecosystem Services
TAC	Total Allowable Catch

LIST OF FIGURES

Figure 1 :Fishing site map where the <i>Decapterus macarellus</i> samples used in this study were fished. The fishing points were mapped based on fishers statements, plotted on Google Earth Pro and the final map created in ArcGIS 9.2	3
Figure 2 : Specimen of <i>Decapterus macarellus</i> (Cuvier, 1833),species.....	8
Figure 3: Data collection at IMar fish laboratory: furcal length (A and B), weight (C) measurement and gonads extraction (D) and weight (E)	11
Figure 4: Social data collection - Survey application to fishery stakeholders (Fish seller and fisherman).	13
Figure 5: Swot analysis: Evaluation of mackerel scad fisheries policies – Minimum Landing Size and Biological Rest Period. The colours on the text represent the statement given by each group (Fishery Institution – green; direct stakeholders (Fishermen) - Blue and Academia & NGO - yellow)	19
Figure 6: Length frequency analyses of <i>D. macarellus</i> between 2003/07 and 2017/21	20
Figure 7: Growth in length comparison between 2003/07 and 2017/21 period.....	21
Figure 8: Length- weight relationship.....	22
Figure 9: Sex ratio comparison between 2003/07 (A) and 2017/21 (B).....	23
Figure 10: Maturity comparison between 2003/07 (A) and 2017/21 (B).	23
Figure 11: Gonad maturation state comparison between 2003/07 (A) and 2017/21(B).....	24
Figure 12: Gonadosomatic Index Comparison between 2003/07 and 2017/21	24
Figure 13: Mature ogive comparison between 2003/07 and 2017/21.....	25
Figure 14: Questionnaire responses percentages about stakeholders' perception and knowledge and about implemented fisheries policies minimum landing size (A) and Biological Rest Periods (B) and responses assessment about Minimum landing size (C) and Biological Rest Periods (D).....	27
Figure 15: Questionnaire responses percentages about stakeholders' perception and evaluation of implemented fisheries policies.....	28

Figure 16: Respondent opinion about laws impact (A) and needed improvements (B) on mackerel scad fishing (word cloud build on R).....	29
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Figure 17: Comparison between large pelagic and <i>Decapterus macarellus</i> landing catch data from 1993 to 2020.....	31
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LIST OF TABLES

Table I: Comparative analysis of the evolution of fisheries policies for <i>D. macarellus</i> between 2008 and 2021(summarized).....	16
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Table II: Questionnaire respondent's characterization according to the groups: Gender Balance, Education Level and Years of Knowledge/experience about fisheries.....	26
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INTRODUCTION

Globally, the rate of fish consumption has increased almost twice as fast as the annual growth rate of the world's population, with a global per capita consumption increasing from 9.0 kg in 1961 to 20.5 kg in 2018, making fish food the most consumed animal protein in 2018 (**FAO, 2020**). According to FAO Code for Responsible Fisheries, fisheries provide a vital source of food, employment, recreation, trade, and economic well-being for people across the world that is why states should promote sustainable fishing to ensure fish food security for present and future generations (**FAO, 1995**).

Due to its diverse importance to humans and the aquatic ecological systems, ensuring sustainable fish production and exploitation, is a step in the right direction to ensure continuous supply (**FAO, 2020**). In Africa, the richest marine resources (**IPBES, 2018**), fisheries are the source of livelihood for 8 million fishers (**Teh et al., 2013; Belhabib et al., 2015**). **Belhabib et al. (2016)** estimated that fisheries GDP could be up to \$20 billion. Particularly, the West Africa region, has the highest fisheries landing catch (**Sambe et al., 2011**) with annual harvest reaching up to 3.5 million tons (**Sambe et al., 2016; FAO, 2020**).

In Cabo Verde, fish is the main food of the population (**PANA II, 2004; González and Tariche, 2009**). Fish consumption per capita increased from 19 kg in 1998, to 26.5 kg in 2004 (**PANA II, 2004; INDP, 2012**). Marine resources are one of the most important natural resources supporting economic activities, and fisheries contribution to GDP varies between 2% and 10% (**DNA, 2015**). Cabo Verde is a small archipelago with a narrow insular shelf of volcanic nature, lack of outcrops and upwelling systems, unfavourable climatic and oceanographic conditions, scarce and irregular precipitation (**INDP 2011b; DNA, 2015**).

This places the ocean as the archipelago's greatest wealth yet, even with marine resources of small abundance and large biodiversity, typically from marine tropical and insular ecosystems. The EEZ of the archipelago covers an area of 734,265 km² (**Bravo de Laguna, 1985**) and more than 99% of area under national jurisdiction is ocean and less than 1% is land (**INDP 2011b; DNA, 2015**).

The fishing potential is estimated between 36,000 and 46,000 tonnes (**PANA II, 2004**) composed mainly by tuna, small pelagic and demersal fish (**Africa Infomarket, 2005**), with high species diversity but low abundances. In order to manage the reduced fisheries resources abundance and species population dynamics, sustainable management laws and measures were implemented (**PANA II, 2004**), disclosed on the decree-law 3/2005 (**República de Cabo Verde, 2005**). Fisheries conservation legal framework and measures are essential for sustainable fish exploitation.

In this regard, the implementation of fisheries policies was used to ensure the adoption of sustainable exploitation of marine resources, in order to prevent fisheries overexploitation, as is the

case for *Decapterus macarellus*, one of the most important commercial and consumed small pelagic fish in Cabo Verde, that, since 2008 has been fished under these policies (**República de Cabo Verde, 2008**).

Despite all the efforts, fisheries in Cabo Verde have been experiencing some difficulties, due to some institutional fragilities such as the need of more research to systematically follow-up and guide fishery legislation due to the lack of funds to hire experts and conduct research (**República de Cabo Verde, 2014**). Moreover, rising resource demand puts more strain on the available natural resources (**Tariche et al., 2011**). Additionally, according to **DeAlteris (2012)**, fishers and reports from Instituto do Mar I.P. (IMar) have been reporting a decline on some fish stock like demersal, small pelagic fish (such as mackerel scad) and crustaceans maybe due to systematic overfishing. In this scenario, we need to know, how effective are the fishing laws policies on mackerel scad stock recovery?

The 5th National Report on the State of Biodiversity in Cabo Verde (**Medina and Gomes, 2015**) and National Biodiversity Strategies and Action Plan (**MAHOT, 2015**) of Cabo Verde reported some improvements in mackerel scad fished under these policies. Some fishers' perception confirmed also that *D. macarellus* catches improved in quantity and quality resulting in higher revenue after some years of low catches. Despite these management improvements, there are still many gaps in the research on fish stock assessment for a global and consistent sustainable management of fisheries in the archipelago.

That is why, this study developed one transdisciplinary research approach methodology, involving state and non-state actors, and will provide baseline data on the perception of fisheries stakeholders on marine fisheries conservation policy and intervention measures, as well as priority adjustment areas for the fishery management. So, the main goal of this study is to evaluate fish stock recovery policy interventions effectiveness on *Decapterus macarellus* fishing. Specifically, by evaluating the effectiveness of: (i) minimum landing size and (ii) biological rest on *D. macarellus* stock recovery, in the north of Cabo Verde, by analysing fisheries management policies to restore the stocks, evaluating the stock's state before and after the implementation of those policies and by (iii) assessing stakeholders' perception about these policies.

The master thesis contains 3 chapters. The first chapter is a Bibliographic Review. The second chapter presents the Material and Methods used and the third chapter devoted to Results and Discussion. The master thesis ends with a conclusion, some perspectives and recommendations. At the end, the bibliographic reference and some annexes and appendices.

CHAPTER 1 – BIBLIOGRAPHIC REVIEW

1.1 Description of the study area

This study aims to undertake an ex-post evaluation of the effectiveness of fishery policy interventions on *Decapterus macarellus* stock, using fish data from the north of Cabo Verde and socio-political analyses on fishery stakeholders' perception about these policies.

Cabo Verde is located in the Northwest Africa (Figure 1), in the dry tropical Sahelian zone characterized by an arid and semi-arid climate, influenced by the Azores anticyclone, the Inter-Tropical Convergence Zone (ITCZ) and the movements of Mid-Atlantic air masses (**Duarte and Romeiras, 2009**).

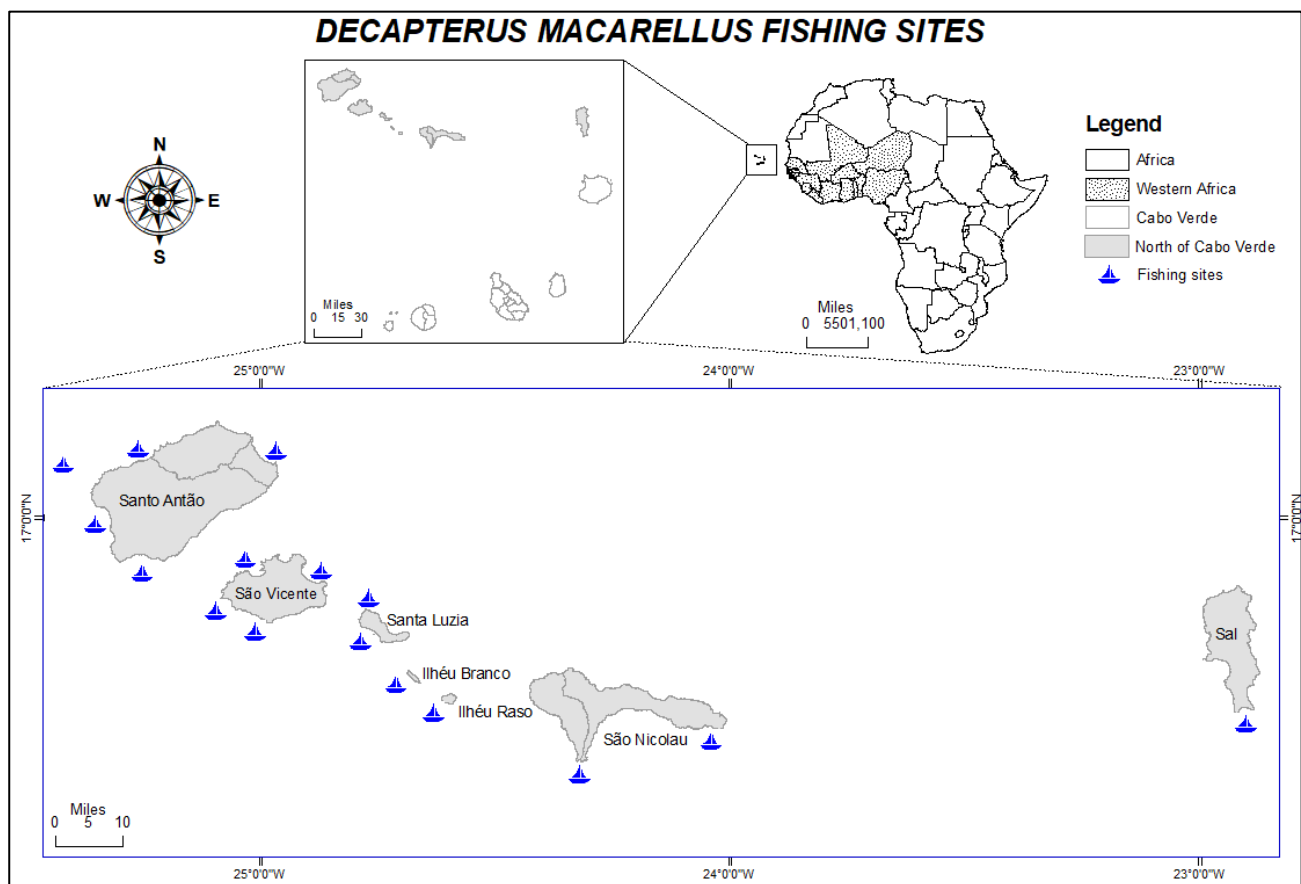


Figure 1 :Fishing site map where the *Decapterus macarellus* samples used in this study were fished. The fishing points were mapped based on fishers statements, plotted on Google Earth Pro and the final map created in ArcGIS 9.2

Moreover, atmospheric and oceanographic patterns, that regulate the seasonal nature of oceanographic phenomena that cause the enrichment of surface waters, low and irregular precipitation, combined with the topographic profile of the archipelago, namely limited extent of the insular shelf and restricted intertidal zone, lead to low population abundance, despite of high marine biodiversity (**MAHOT, 2014**). Therefore, the annual rainfall season mainly occurs from August to October reaching 122.186 mm (mean $\pm$ SD of 148.5 ± 15.9 mm) (**Hijmans et al., 2005**), and the evaporation exceeds the precipitation (**Duarte and Romeiras, 2009**). The climate is moderated by

the ocean and the north-eastern trade winds that carry moderate humidity throughout the year (**Hijmans *et al.*, 2005**).

The mean annual temperature varies from 22.2 to 23.8° C (mean 23.1 ± 0.5° C), but during the warmest period can reach around 26.7 to 28.1° C (± 0.4° C) and during the coolest month decreases, varying from 18.0 to 19.4° C (± 0.4° C) (**Hijmans *et al.*, 2005**). The sea surface temperature (SST) in the northern part is colder than the southern part of the country, an average 24 °C. Besides, the SST anomaly's coolest season goes from January to June where the coldest months are February and March, and the warmest season goes from July to December, where September and October are the warmest months (**Vieira, 2018**).

1.2 Management fisheries in Cabo Verde

In Cabo Verde, fish and fish products are the main food of the population, with the fish consumption per capita increasing from 19 kg in 1998, to 26.5 kg in 2004 (**PANA II, 2004; INDP, 2012; DNA, 2015**). Aside, there is no available data, updated at a local scale, and the National Report on the state of Biodiversity in Cabo Verde, in 2015, assumes that these levels of production/consumption of fish and fishery products have been pointed out as the main factors supporting population growth in Cabo Verde (**DNA, 2015**). Marine resources are one of the most important natural resources supporting economic activities, and fisheries contribution to GDP varies between 2% and 10% (**DNA, 2015**). This makes the archipelago's greatest wealth yet, even with marine resources abundance and biodiversity typically from marine tropical and insular ecosystems. Cabo Verde is a small archipelago with a narrow insular shelf of volcanic nature, lack of outcrops and upwelling systems, unfavourable climatic and oceanographic conditions, scarce and irregular precipitation (**INDP 2011b; DNA, 2015**).

Its fishing potential is estimated between 36,000 and 46,000 tonnes composed mainly by tuna, small pelagic and demersal fish (**PANA II, 2004**). In order to manage the reduced fisheries resources abundance and species population dynamics, sustainable management and measures were implemented guaranteeing species perpetuation (**Tariche and Martins, 2006**). That is why, fisheries management in Cabo Verde has been made up of three interconnected and interdependent systems since the beginning, each of which is accountable for the effectiveness of fishery management while also considering social, economic, and biological elements of fishing.

The fisheries management system in Cabo Verde includes management (administration-related research), political/legal (legislation and regulation), and enforcement (monitoring, control, and surveillance) roles that are performed by various institutions within the specified domains (**INDP, 2015; República de Cabo Verde, 2021**). Thus, according to the official bulletin of Cabo Verde Republic, I Series - N° 142 of December 31, 2020, various types of fishing are practised in the

Cabo Verde EEZ. Each fishery is managed according to the target species, as well as the fishing methods, to assess and manage it in its current exploitation level, with the goal of preserving the species and the fishery activity. Among the different fisheries, the following ones can be described:

- Small-scale fishing is the most traditional fishery and is performed within the 3 nautical miles, representing an important source of employment and income generation for most coastal communities. It is most performed by full-time fishers but, during years of low rainfall and consequently low agricultural activity, part-time fishers also practice it. The target species are tuna and demersal fishes with handling, small pelagic fishes (mackerel scad and horse mackerel) with purse seine, guide loop and bent with gillnet. It is also reported coastal lobsters with creels and, demersal fish, common whelk and other molluscs with snorkelling.
- Semi-industrial/industrial fisheries, the most exploratory fisheries done by nationals, performed between 3 and 12 nautical miles from the coast, targeting tuna fishes (yellowfin, skipjack and frigate tuna) with line and heel, pole or purse seine, small pelagic fishes (mackerel scad and big-eye) with purse seines. It is also reported pink lobster and deep-sea crab with creels, sharks, and shrimp.
- Foreign industrial fishing, allowed beyond the 12 nautical miles, for example on the bilateral fishing agreements between Cabo Verde and the European Union.
- Amateur fishing for recreational, entertainment, tourist or sports purposes
- Live bait fishing for pole-and-line tuna fishing purposes.
- Sea cucumbers fishing by diving.

Fisheries in Cabo Verde have been experiencing some difficulties, due to some institutional fragilities such as the need of more research to systematically follow-up and guide fishery legislation, also due to the lack of funds to hire experts and conduct research (**República de Cabo Verde, 2014**). Moreover, rising resource demand puts more strain on the available natural resources (**Tariche et al., 2011**). Additionally, according to **DeAlteris (2012)**, fishers and reports from Instituto do Mar I.P. (IMar) have been reporting a decline on some fish stock like demersal, small pelagic fish (such as mackerel scad) and crustaceans maybe due to systematic overfishing.

Fisheries management and action plans are updated throughout time to ensure that sustainable measures are in place to regulate and encourage sustainable exploitation. The decree-law 53/2005, which guided by catch assessments, established steps to govern fisheries in a sustainable way by enhancing fishing management, reducing post-harvest loss, and maintaining ecosystem productivity (**República de Cabo Verde, 2005**).

More research led to the establishment of more sustainable regulations in 2008, namely the biological rest periods and minimum landing size for mackerel scad and other species. A long process required ecological studies, social consultancy, policy formulation followed by long sessions of socio-economic and political consultation among representatives of fisheries stakeholders. Nowadays, the current implemented policies are disclosed with updated, approved and implemented policies for fish stock restoration such as:

- Control of nominal fishing effort;
- Establishment of a total allowable catch (TAC);
- Allocation of individual production quotas;
- Minimum landing size;
- Implementation of closed seasons or biological rest periods;

Nine of the commercial species of Cabo Verde are actually fished under this policies: *Decapterus macarellus* (mackerel scad), *Spicara melanurus* (Blackspot picarel), *Selar crumenophthalmus* (Big-eye scad), *Cephalopholis teaniops* (Spotted grouper), *Lithognathus mormyrus* (Sand steenbras), *Palinurus charlestoni* (Spiny pink lobster), *Panulirus echinatus* (Spiny brown lobster), *Panulirus regius* (Spiny green lobster) and *Scyllarides latus* (Slipper lobster), *Auxis thazard* (frigate tuna) and *Auxis rochei* (bullet tuna). All these policies work together to ensure sufficient reproductive fish biomass (**King, 1995**), and prevent overexploitation related to fish growth and recruitment for the fish population, as well as for the global fisheries (**Hill, 1992**).

Despite these management improvements, there are still many gaps in marine research and fish stock assessment for a global and consistent sustainable management of fisheries in the archipelago. Fish stock assessments have long been used as tools in fisheries management, particularly now that the world is facing global overfishing (**FAO, 2020**). It is fundamental to contribute and ensure the sustainable development of the fisheries sector and fisheries resources exploitation, especially in the case of Cabo Verde, because nonetheless a big EEZ of known little marine resources abundance, fisheries provide an important contribution to GDP (**DNA, 2015**).

In such a context, to keep fishery resources exploited in a sustainable manner, it is necessary to follow up the implemented laws, through ex-post evaluation assessment, in order to evaluate the effectiveness of such measures, improve and reformulate the fishery legal framework. Fish conservation legislations and measures are essential for sustainable fish exploitation. In this regard, the implementation of fisheries policies are used in Cabo Verde to ensure the adoption of sustainable exploitation of marine resources, in order to preserve species from being overexploited, as is the case

for *D. macarellus*, one of the most important commercial and small pelagic fish in Cabo Verde (**Tariche et al., 2011**).

The mackerel scad fishery is vital for the Caboverdean economy and food security, accessible to all population, delicious, nutritious and a low-cost fish. Juveniles are used as bait in the large pelagic fisheries (**Santos et al. 2012**). Official landing catch data from INDP during the period from 1989 to 2015 indicates that mackerel scad made up almost 40% of Cabo Verdean total catches, with highest landing catches in 1997 and 1998 (Figure 16). After this peak the catch has decreased significantly, representing only 6.6 % of the landing in 2015 (**Vieira, 2018**). Fishers complain that mackerel scad fishing became difficult, as they need to go far more than usual to fish it. IMAR research reports under the Overseas Fishery Cooperation Foundation Project (**MAAP, 2004**), **Tariche and Martins (2006)** and **Stobberup and Erzini (2006)** have shown that mackerel scad abundance was decreasing.

DeAlteris (2012) reported mackerel scad potential stock of 2700 metric tons per year, while previous assessments pointed between 4000 to 6000 metric tons (**Stobberup and Erzini, 2006**). Mackerel scad landing catches have been shown in an alarming decline (**Martins, 2014**). This could be due to excessive fishing effort and adverse environmental conditions, surely bringing negative consequences that could be, practically irreversible, with major socio-economic impacts (**Tariche et al., 2011**).

Earlier studies from **Almada (1997)**, **Carvalho and Caramelo (1999)** and **Jardim (1999)** also reported that the increasing of mackerel scad landing catch during the 90s were related to the increasing of fishing effort caused by two Japanese freezing vessels which have been buying mackerel scad from small-scale fishing to use as large pelagic fish bait. Additionally, the bigger prices offered by the Japanese and the introduction of small semi-industrial purse seiners contributed to increase the mackerel scad landing catch. Also, with the opening of a canning processing plant in 2000, the demand of mackerel scad increased (**Vieira, 2018**).

Consequently, fishing effort increased, from 2004 to 2006 (increasing of fishing permits and introduction of bigger and more sophisticated vessels) the landing catch increased in the same period (**Tariche et al., 2011**). Several reasons might cause the fluctuations on mackerel scad landing catch, such as the distribution pattern off the shelf, potential migration of mackerel scad stock between islands (**Ramos and Fujiwara, 2004**), sea surface temperature oscillations and primary productivity (**Vieira, 2018**).

Additionally, during the spawning period mackerel scad is more vulnerable, and this were corresponding to the high catches period, consequently, fishers were fishing higher quantities than

the vessels storage capacity, resulting in a huge loss of biomass (**Tariche et al., 2011**). Also increases of population causes increases on fish demand, consequently overfishing can be occurring over the last 30 years (**DeAlteris, 2012**) to supply the market demand.

All of these factors combined with insular marine ecosystem limitations on the Cabo Verde EEZ can be causing the oscillations on *D. macarellus* landing catch. Consequently, in this context, all these factors pushed the fisheries administration and management system to the definition and implementation of biological rest periods and minimum landing catch policies. The policies have been updated according to outputs from few studies related to mackerel scad using different methods (**Fortes, 2019**) and fishers' observations and complains.

However, no studies have assessed the effectiveness of these policies and thus the fisheries administration have limited scientific or management elements that can allow these measures to be adjusted according to the new and successive dynamic balances that are being established in the fishing activity as a whole. Because of this, research employing flexible methodologies are needed, so that Cabo Verde fisheries management can know what progress has been made, as well as the rules and policies that need to be updated in order to reformulate his fishing law policy.

1.3 *Decapterus macarellus* (Cuvier, 1833)

1.3.1 Description

Decapterus macarellus (**Cuvier, 1833**), commonly known as mackerel scad or “cavala preta” by caboverdean (Figure 2), belongs to the family Carangidae. The species is characterized by having an elongated body, of proximally rounded cross-section with a metallic blue and blue-grey colouration on the dorsal side, the first dorsal fin with 7 hard rays and the second with 1 thorn and 31 to 37 soft rays. Anal with 2 separate thorns, followed by 1 thorn and 27 to 31 soft rays. Curved part of the lateral line with 68 to 79 scales. Branchlets numbering 9 to 13 on the upper branch and 31 to 39 on the lower branch of the first branchial arch (**Reiner, 2014**).



Figure 2 : Specimen of *Decapterus macarellus* (Cuvier, 1833), species

1.3.2 Biology

The mackerel scad has relatively moderate growth. It can reach its maximum size at 7 years of age, in Cabo Verde (**Almada, 1997**). The largest specimen recorded in Cabo Verde has 39 cm of furcal length (Santos, 2018), but it can grow up to 20 cm in one year (**Honebrink, 2000**) with a bigger growth rate during the summer (**Shiraishi *et al.*, 2010**). Its spawning season is in the hottest periods of the year, from April to October with higher intensity in summer (**Tariche and Martins, 2006**). During this period, the mackerel scad moves from the coast to deeper waters (**Tariche *et al.*, 2011**).

1.3.3 Ecology

It presents a circumpolar distribution from tropical to subtropical seas, mostly found on clear oceanic waters, and is widely associated with oceanic islands between 0-400m depths but commonly found in depths of 40-200m (**Froese and Pauly, 2018**). The individuals feed, during the night, mainly on copepods and decapod larvae (**McNaughton, 2008**), and adults feed mostly on larger crustaceans from the order Euphausiacea (**Almada, 1997**). In addition, large pelagic fishes, like tuna species and some seabirds (**McNaughton, 2008**), predate on it.

CHAPTER 2 - MATERIAL AND METHODS

2.1. Material

2.1.1. Biological material

All specimens examined in the present study belong to the species *Decapterus macarellus* (Figure 2) fished in northern islands of Cabo Verde (Figure 1).

2.1.2. Technical material

During the ecological and social data collection, the following materials were used to facilitate the sampling:

- Thermic sampling box;
- Ictiometer;
- Electronic balance;
- Dissection kit: scissors, tweezers and knife;
- Maturation states guide;
- Gloves;
- Gown;
- Petri dish;
- Tablet;
- Sampling formulary;

2.2. Field study methodology

A transdisciplinary study inspired by IPBES conceptual framework – connecting nature to people developed by **Díaz *et al.* (2015)** was carried out on the analysis of the legislation regulating the fishing of mackerel scad. We evaluated the stock in two periods, before (from 2003 to 2007) and after (from 2017 to 2021) the implementation of fishing policies, and the fishing stakeholders' perception. The methods were adjusted from "Ex-post" analysis (**OECD, 2016**), stock assessment (**Sparre and Venema, 1997; Cadima, 2000**) and adapted socio-political studies (**Brown *et al.*, 2019; Kelemen *et al.*, 2021**), in order to answer the study objectives.

2.2.1 Policy analysis method

Fish stock restoration policies on the Fisheries Resource Management Plan, disclosed on the official bulletin series, since the law's implementation in 2008, have been assessed. Additionally, to complement the policy analysis, and underline needed adjustments of these policies, one SWOT

(Strengths, Weaknesses, Opportunities, and Threats) analysis (adapted from seesionlab.com) was undertaken during a fishery stakeholders' workshop organised on October 15, 2021;

This workshop was held with the objective of analysing fisheries management policies for restoring the *D. macarellus* stock, more precisely BRP and MLS. Representative members of fisheries stakeholder groups, such as fishers, fisheries institutions, academia, and non-governmental organizations (NGOs) dealing with the regulations, were the target audience. Participants were divided into four groups depending on their function in the fishing industry. Each group created a SWOT analysis based on their understanding and viewpoints on the policies (due to unforeseen circumstances, some fisheries institutions were unable to participate, and so the NGOs and academia representatives were combined in one group). During the performance of their work, each group was aided and guided. Following the working group, each one presented their results and a considerable debate ensued.

2.2.2 Ecological methods for sampling and analyses

During three months (August to October) of commercial sampling, 225 random samples of *D. macarellus* were bought at the fish market in São Vicente. Questionnaire was applied to fishers on the market and landing port of Complexo de Pesca de Cova d'Inglesa to get all information about the fishing sites, among others.

After, the samples were taken to IMar fish lab for further data collection. Each individual was measured the furcal length using an ictiometer, weighted using an electronic scale, and. At the end, the gonads were collected, weighted and the maturation state were noted (Figure 3).



Figure 3: Data collection at IMar fish laboratory: furcal length (A and B), weight (C) measurement and gonads extraction (D) and weight (E) .

Data collected from 2003 up to July 2021 (n=7213), provided by IMar were added to the new collected data. The data were separated in two series for comparative analysis: 2003 to 2008 and 2017 to 2021, respectively before and after the implementation of the management policies.

Ecological analyses we used were computed by using Microsoft Excel, FiSAT II and R software (RStudio, 2020), based on fish stock assessment methods proposed by **Sparre and Venema (1997)**, **Cadima (2000)**, **Tariche and Martins (2006)** and **Santos (2018)**, to assess *D. macarellus* stock. The following models and equations were used, and the results were statistically tested at significance level of 5%:

- Length frequency distribution was graphically compared in the considered temporal scale and statistically tested by *t* student test.
- Sex ratio were compared graphically between the time series and statistically tested using Kruskal-Wallis' test.
- Growth in length were analysed using the **Von Bertalanffy (1938)** model, adapted by **Beverton and Holt (1956)**:

$$L(t) = L_{\infty} \times [1 - e^{-K \times (t-t_0)}]$$

where $L(t)$ is the length at age t , L_{∞} is the theoretical or asymptotic length that individuals can reach if they live indefinitely. K is the growth coefficient that indicates the speed with which L_{∞} is reached (**Francis, 1986; King, 1995**). T is the age and t_0 is the theoretical age when length equals zero. Growth parameters (L_{∞} and K) were estimated using *FiSAT II* with $t = t_0$ is estimated using **Pauly (1997)** model:

$$(-t_0) = -0.392 - 0.275 \log L_{\infty} - 1.038 \log K.$$

- Gonad maturation were analysed based on the gonad maturation states guide (annex I) and distributed in a column chart by month. The gonadosomatic Index (GSI) were calculated as follow (**King, 1995**):

$$GSI = 100 \times \left(\frac{GW}{TW} \right)$$

GW is the gonad weight and TW is the total weight of the fish in grams. These two results were used to filter sexually mature individuals for the maturation ogive construction using **Haddon (2001)** logistic model:

$$P = \frac{1}{1 + e^{\frac{-(l-l_{50})}{\delta}}}$$

where P is the proportion of mature fish per length class, l , l_{50} and δ are the model parameters, l_{50} is the size at first sexual maturity or length at which 50% of individuals are sexually mature.

2.2.3 Social method to assess stakeholders' perception

An online social survey, created on google forms, was conducted based on stakeholder's analysis structure method (adapted from www.sessionlab.com), to understand fishery stakeholder's perception about the implemented policies for *D. macarellus* stock recovery, taking into account the gender balance. To comply with COVID-19 social policy, links to questionnaires were sent to relevant stakeholders (n=100) at fishery institutions, non-governmental organizations and academia and shared to civil society (consumers) on Facebook.

The same questionnaire was directed face-to-face in the landing ports and fish markets and São Pedro and Salamasa villages, in São Vicente (Figure 4), always maintaining safety measures related to COVID-19, to stakeholders (n= 75) who did not have access to online communication channels, respecting COVID-19 restrictions. After, the social data were analysed using statistical descriptive methods and tests on Excel and SPSS Statistics software.



Figure 4: Social data collection - Survey application to fishery stakeholders (Fish seller and fisherman).

CHAPTER 3 - RESULTS AND DISCUSSION

3.1 Fishery policies analysis

Since colonial times, Cabo Verde has been implementing laws for the management of the most vulnerable fisheries resources, as in the case of Decree No. 7/1879. Besides, it was after the ratification of the United Nations Convention on the Law of the Sea - UNCLOS (Law No. 17/III/87) that the creation of laws for the sustainable management of fisheries intensified. Consequently, in 1987, the Decree-Law No. 17/87 defined the general principles of the policy for the exploitation of fisheries resources. In Addition, the Decree No. 97/87 established the standards of protection, conservation measures and inspection, among other laws and policies related to the sea and marine resource's export (**INDP, 2011b**).

Thus, since earlier times (1990), the management of caboverdean fisheries is based on a system based on 3 independent pillars: Fisheries Research (IMar), Fisheries Administration (DNPA) and Inspection (IGP, IMP, CG, MP and NP). IMar is responsible for producing the management recommendations, based on fisheries research and complementary studies such as oceanographic and socio-economic research. These studies are presented to the Fishery Administration for subsequent formulation and implementation of fisheries laws. The Inspection body is responsible for ensuring that the laws are respected and complied by the other fishing stakeholders (**PANA II, 2004; INDP, 2004**).

Management measures are being updated, but few changes have been made since the law's implementation, based on observations and few studies (Table I). Currently, the valid management measures are presented in the General Fisheries Resources Plan (with 4 years of validation) which contains the development and management measures for different fisheries. The Plan contains the policies to guide, in a sustainable manner, and to ensure rational and efficient economic exploitation of resources, conservation, protection, recovery of marine and coastal ecosystems, maintenance of good environmental status, optimisation of catches and adding value to fish and fisheries products, focusing on promoting competitiveness and increasing the contribution of fishing to the national economy (**PANA II, 2004; INDP, 2011**).

Aligned with the Government Programme, the Strategic Plan for Sustainable Development (**PEDS 2017-2021**) and in the framework of the objectives of the Promotion of Blue Growth, the main objective is to promote the improvement of the fisheries sector. In addition, ensure its cautious development in order to provide maximum sustainable economic results, adding value to fisheries

products (transformation and commercialisation), thus contributing to the reduction of the balance of payments deficit and to food security (**Republica de Cabo Verde, 2020**).

The management measures are implemented through Executive Plans, first biannual (2005 to 2017) and now, annually (2018 to 2021) and published in the Official Bulletin of the Republic of Cabo Verde (accessible in the website <https://kiosk.incv.cv/>), containing mandatory compliance measures, subject to monitoring and the power to sanction in case of contravention (**INDP, 2011b; República de Cabo Verde, 2021**).

Among the main measures contained in the plan are: (i) the laws on areas reserved for fishing, (ii) the prohibition of harmful fishing practices, (iii) the establishment of minimum sizes of target species and (iv) the establishment of closed seasons for endangered species (**Republica de Cabo Verde, 2021**).

The implementation of these laws for *D. macarellus* caused a lot of polemic discussion among the stakeholders because it is reported as one of the most explored small pelagic fish since the beginning of fisheries studies in Cabo Verde (**Almada, 1997**) with high nutritive value and monetary accessible to everybody (**INDP, 2007**). Therefore, **DeAlteris (2012)** reported that *D. macarellus* have been overfished. The current valid laws are disclosed in the Resolution No. 63/2021 from the Serie I – n° 142 Sup. «B.O.» of the Republic of Cabo Verde - December 31, 2020 (Table I).

The resource is fished by an industrial purse seiner of more than 8 meters long and by small-scale vessels of 8 to 10 meters long, with an outboard motor. They are multi species fishing directed vessels based on species availability and they must follow the following management:

- This fishery is reserved for national vessels and controlled by fishing licences;
- Establishment of a closed season from July 15th to September 14th (but since 2020 is having an exceptional suspension due the actual Covid-19 pandemic.
- Establishment of the minimum catch and commercial size at 20 cm (furcal length). For live bait fishing purposes, the minimum size is 6 cm.

Table I: Comparative analysis of the evolution of fisheries policies for *D. macarellus* between 2008 and 2021(summarized)

Resolution No. 11/2007	Resolution No. 10/2009	Resolution No. 56/2014	Resolution No. 1/2016	Resolution No. 29/2016	Resolution No. 8/2018	Resolution No. 93/2020	Resolution No. 63/2021
2007 - 2008	2009 - 2010	2014 - 2015	2016	2016 - 2017	2018	2020	2021
Authorise the fishery by industrial purse seiner, usually multi-purpose, depending on fish species availability.							
This fishery is reserved for national vessels.							
Cautious expansion of fishing effort through fishing licences control							
Allowed for purse seiner bigger than 6.5 metres in length.		Allowed for purse seiner bigger than 8 metres in length.				Allowed for purse seiner between 8 and 10 metres, with outboard motor.	
Establishment of Biological rest period between August 1 to September 30, starting on January 2008				Establishment of biological rest period, between July 15 and September 14.		Exceptional suspension of biological rest period, (due to COVID-19 pandemic)	
-	Margin of tolerance allowed 10% of the total weight of the catch in the case of by-catches of mackerel scad in the purse seine fishery.						
Establishment of minimum landing size, to be marketed, of 18 cm furcal length, starting in January 2008.				Establishment of minimum landing size, to be marketed, of 20 cm focal length.			
-	Ban on the marketing of mackerel scad caught on the previous terms, except when used as live bait. The minimum size for bait is kept at 6 cm (small-scale fishing)						
-	Authorisation of semi-industrial seine fishing within 3 nautical miles exclusively for the capture of small pelagic, except in all bays of the archipelago						

In 2020, Covid-19 restrictions imposed by the Cabo Verde government, to contain the spread of the virus, led to an economic crisis caused by home confinement. As a result, *D. macarellus* closed season policy has been suspended at least until 2021, as the fishing sector faces significant indirect impacts of pandemic restrictions, and the state cannot provide subsidies to fishing operators (República de Cabo Verde ,2021).

3.1.2 SWOT analysis of *D. macarellus* fishing management policies

Twenty fishery stakeholders attended the workshop. They easily collaborated on the methodology applied during the workshop. The results of the group and plenary discussions were summarised in Figure 5. The fisheries institutions group focused mainly on the positive points, the fishers group on the weak points and the academia and NGOs group on the opportunities that the implemented policies provide for improving the sustainable management of the resource. However, it was observed that the groups share common ideas, as they both agree that the policies are beneficial for the mackerel scad fishery and that certain reformulations need to minimize the negative points, such as the improvement of social consultation, better enforcement and monitoring of the laws.

During the plenary session, the stakeholders had the opportunity to discuss their views on the policies. The fishers did not miss the opportunity to question, “*where is the mackerel scad?*”, “*Why has it disappeared from the catches?*” However, they themselves believe that mackerel scad didn’t disappear, but it is migrating as the mackerel scad started to appear in the bottom of the net and in places further away from the shore. So, when the bigger boats increased the nets and started to move further away from the shore, the mackerel scad reappeared in the catches.

However, the catches have been reduced lately. Fishers, while agreeing with the law's implementation, are dissatisfied with it because they are primarily concerned with financial returns and believe that the law is to blame for a decrease in catches at a time when they are most abundant, causing them to form larger schools and migrate to other areas. Because the mackerel scad spawns during all year, the closed season should be properly adjusted for them, as the closed season does not appear to be as effective on the stated dates.

The group of fishing institutions responded to their concerns by stating that, for the implementation of the laws, the social, economic and biological aspects of the mackerel scad must be taken into account. Therefore, the implemented measures were the result of meetings held with stakeholders, where IMAR presented the results of its studies and that these were concurrent with the other fishing institutions, in the presence of the minister and representatives of the stakeholders.

Together they agreed on the best way to define laws that would minimise the consequences for all stakeholders involved in the process. They also clarified that the reasons behind the disappearance of mackerel scad in the catch is still unknown, but may be related to climate change that impacts ocean temperatures, food availability and alters the natural habitat of mackerel scad. Additionally, they drew attention to the fact that the increase in population also influences the pressure and exploitation of the resource and makes management difficult.

Fishers voiced the lack of government support during the closed season, claiming that they rely solely on fishing and that this results in a loss of income. A monetary incentive at this time could assist them in complying with the law. On the other hand, the group of fishing institutions maintains that there are no finances available to create this incentive. However, they are conducting research and projects to try to figure out why black mackerel scad are disappearing from the catch. Meanwhile, they have encouraged fishers to exploit certain less commonly caught species, as a shortage of fish allows fishers to consume crab, which creates revenue, ensures the fishers' livelihood, and guarantees an alternative and healthy source of protein for the population.

Even if the socio-economic impact can be negative, due to income generation reduction, these vessels have alternative fisheries during this period, like big-eye tuna, which is open to fishing. Also, this period coincides with the high season for tuna species fishing, which is more profitable as they are bigger than mackerel scad and they are mostly sold to fish processing industries (**República de Cabo Verde, 2021**).

The group of academia and NGOs, which focused mainly on opportunities, will highlight the importance of developing a fund to hire more experts. As many are forced to flee to other areas as they are not available in that field, thus leading to a loss of qualified experts with the capacity to improve sustainable management on the basis of sustainable development goals.

SWOT ANALISYS

Evaluation of mackerel scad fisheries policies: Minimum Landing Size and Biological Rest Periods

STRENGTHS

- Existence of management measures for sustainable fisheries management;
- Reduction of fishing effort and impact on fishing, causing an increase in mackerel scad stock and a decrease in fishing mortality;
- Protection of mackerel scad spawning (BRP), aiming specie prosperity;
- Inclusive implementation of fisheries mangament measures with social consultation;
- Reduction of unnecessary fishing mortality due BRP
- Promotion of sustainable fishing;
- Exploitation of under-exploited and/or little-known species;
- Increase in LMS;

WEAKNESSES

- MLS and BRP are not properly adjusted to specificity of the species;
- Ineffective social consultation on the policymaking process;
- Direct stakeholders income generation reduction due BRP;
- Poor involvement of all stakeholders in policy making;
- Weak laws dissemination
- Lack of experts involvement in policies formulation/monitorazation;
- Lack of stock assessment studies on mackerel scad;
- Inefficient enforcement
- Inexistence of monitoring of implemented laws

OPPORTUNITIES

- Create condition for more studies that allows periodically laws reformulation;
- Studies to understand the causes of mackerel disappearance on catches;
- Improve and increase the fishing effort (bigger and more sophisticated vessels;
- Development/strengthening of partnerships with NGOs and fishing communities (improve social consultation);
- Improve and increase the dissemination of laws;
- Inclusion and greater valorisation of fishermen's ILK;
- Development of actions to monitor and reformulate the laws;
- Stakeholders involvement in creation/reformulation of fishing laws;

TREATS

- MLS below mackerel scad maturation size may comprise a stock prosperity;
- Climate change cause sea surface temperature changes, consequently mackerel habitat changes, pushing them to migrate for better habitats;
- Overfishing of big pelagic (mackerel predators) allows mackerel scad to migrate;
- Lack of knowledge about the laws for most of population;
- Lack of funds to hire experts;
- Inexistence of alternative activities during BRP;
- Lack of monitoring and ex-post evaluation;
- Inefficient inspection efforts;

Legend: Fishermen, Fishery institutions, Academia and NGO , All together

Figure 5: Swot analysis: Evaluation of mackerel scad fisheries policies – Minimum Landing Size and Biological Rest Period. The colours on the text represent the statement given by eacheach group (Fishery Institution – green; direct stakeholders (Fishermen) - Blue and Academia & NGO - yellow)

3.2 *Decapterus macarellus*' stock state before and after the implementation of the fishing management policies

Results from data analysis of 7,438 individuals, distributed in two groups, before (2003 to 2007) and after (2017 to 2021) the implementation of the restoration policies for the species under analysis revealed that the fishing under the implemented policy of Biological Rest Periods (BRP) and Minimum Landing Size (MLS) caused some changes on mackerel scad fishing.

3.2.1 *Decapterus macarellus* length frequency comparison

Length frequency analyses of *D. macarellus* showed that the sampled individuals ranged from 16 to 35 centimetres in the first period (2003-2007) and 18 to 45 centimetres in the second period (2017-2021). In other words, there was an increase in the average of landing catch size of mackerel scad and in the frequency of larger mackerel scad landing catch, as shown in the graph below by the displacement and flattening of the normality curve of the length frequency (Figure 6).

Similar results were obtained by **Santos (2018)** where the furcal length was bigger, in comparison of **Jardim (1999)** assessment, but this study sampled bigger mackerel scad than in **Santos (2018)**.

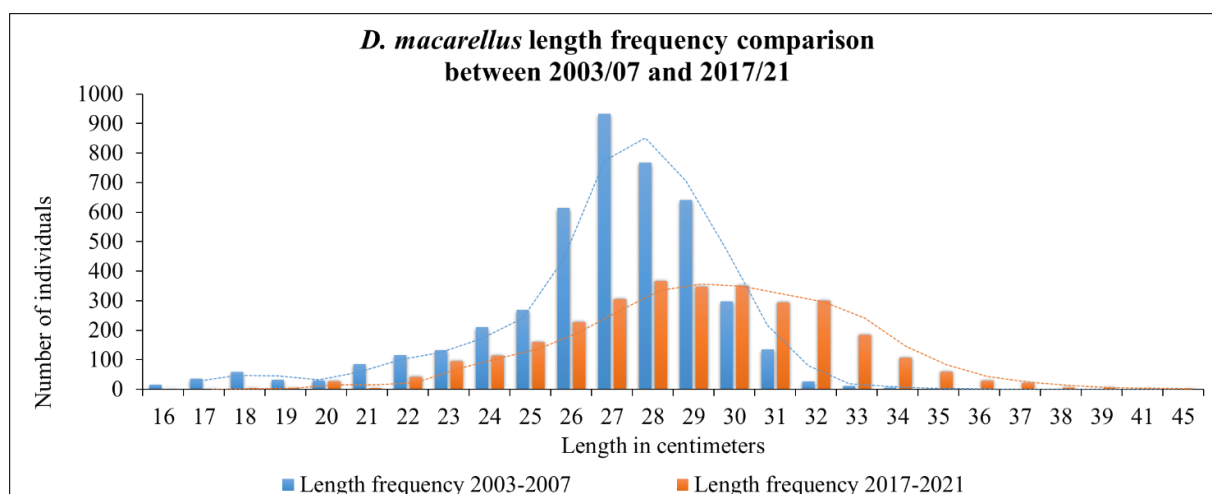


Figure 6: Length frequency analyses of *D. macarellus* between 2003/07 and 2017/21

This indicates that the implemented laws MLS and BRP, have made statistically significant changes on mackerel scad catches, tested by *t*-student test (*p*-value <0.05). The biggest mackerel scad reported on IMAR database, were caught during October sampling. This result indicates that mackerel scad is growing more before the recruitment for fishing.

3.2.3 Growth in length comparison

The growth parameters were higher in the second period 2017/21 ($L_{\infty} = 46.73$ cm FL; $K=0.47$ year⁻¹) than in the first period 2003/07 ($L_{\infty} = 41.48$; $K=0.39$ year⁻¹) (Figure 7). Similar values were

found by **Almada (1997)** ($L_{\infty} = 30.1$ FL and $K=0.34$ year⁻¹) for the first period in analysis. Also, **Santos (2018)** ($L_{\infty} = 41.83$ cm FL and $K= 0.39$ year⁻¹) and **Vieira (2018)** ($L_{\infty} = 40.6$ cm FL and $K= 0.45$ year⁻¹) similar to the second period in analysis in this study, using the same data source, but different periods of analysis.

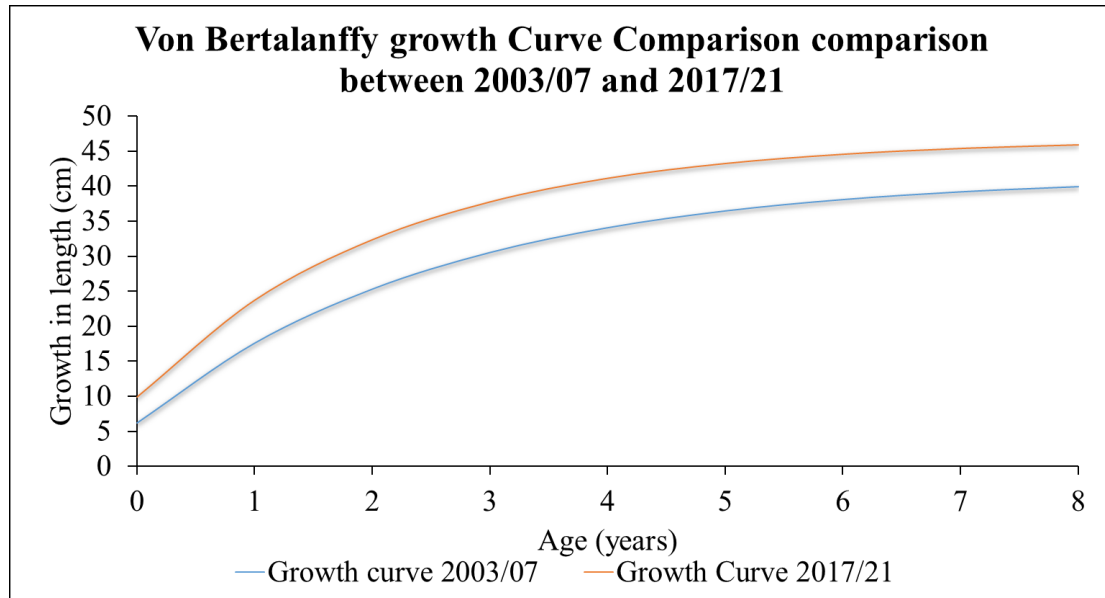


Figure 7: Growth in length comparison between 2003/07 and 2017/21 period

The mackerel landing size is growing more before being fished than in the first period. According to **Beverton and Holt (1957)**, growth can be affected by several factors, biotic or abiotic, such as food availability, population density, habitat changes and sea temperature (**Beverton & Holt, 1956**).

3.2.4 Length-weight relationship

The length-weight relationship is strongly correlated in both periods, with $R^2 = 0.9064$ in the 2003/07 period and $R^2 = 0.9337$ in 2017/21 (Figure 8). For the first period and second period, the allometry coefficient was $b_1 = 3.14$ and $b_2 = 3.30$ and the condition factor $q_1 = 0.0082$ and $q_2 = 0.0044$ consecutively.

Similar values of both correlation factor, allometry coefficient and condition factor were found by **Almada (1997)**, **Jardim (1999)**, **Vieira (2018)** that means that mackerel scad have positive allometric growth but is not isometric, while **Santos (2018)** results shown and isometric growth for mackerel scad $b=3.01$. To have an isometric growth, meaning the fish have growth in length and weight percentages and proportional growth, the species should have at least $b=3$ (**Cadima, 2000**).

Even if the allometry coefficient were close to this value, we cannot confirm that mackerel have an isometric growth, because various factors can influence its growth in length and weight. For example, **Almada (1997)** and **Jardim (1999)** used less data to determine mackerel growth parameters

and the samples had under 33cm of furcal length. **Vieira (2018)**, analysed data from 1990 to 2015 as one group, including the smallest and biggest length values registered on IMar database.

Santos (2018) analysed the data from 2005 up to 2018, consequently smaller and bigger mackerel scad, up to 39cm of furcal length. In this work for instance, the data were separated and analysed on 2 periods, smaller mackerel length frequency were analysed in the first period before the implementation, while bigger mackerel, up to 45 cm were analysed in the second period, showing that mackerel don't have isometric growth as mention for the others authors.

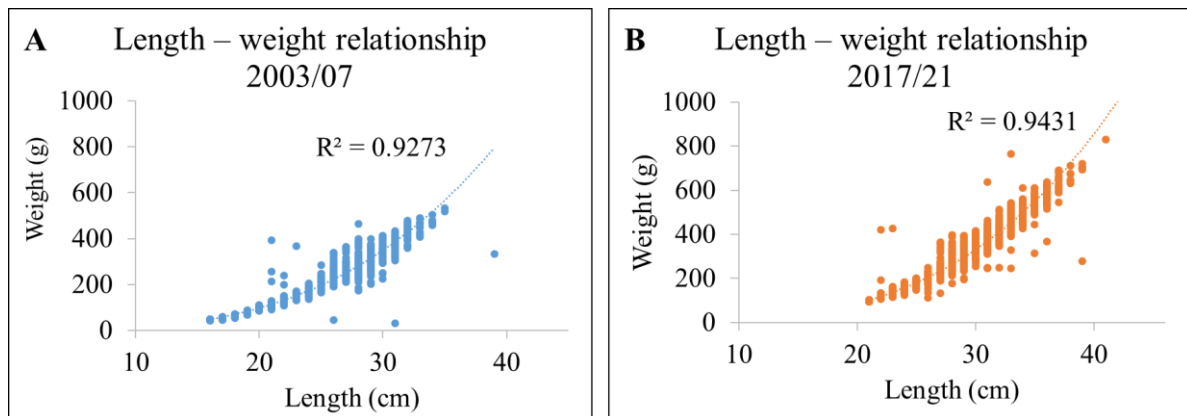


Figure 8: Length- weight relationship

This result shows that the sampling size and fish length/weight are influencing the results. In the first period smaller values of allometric factor were observed (Figure 8). Despite this, other different factors could be behind these results, such as the biotic and abiotic conditions of the day they were fished. For example, the weather season, their life cycle and sex (if reproducing or not, juvenile, male or female), the habitat (if is favourable or not, if they are in their migrating period or not), gonad maturity (if they are ripped or not). Also, the diet and time that they have eaten before being fished, also their health condition.

Moreover, also the preservation measures, like mackerel scad case in this study, the differences in the length and weight ranges that they have been caught affect this relationship. According to **Martin *et al.* (2016)**, the length-weight relationship differs among species, related to their body shape and individual fish conditions like food availability and growth, and all of this are variable and dynamic. Also, in the same sample, fish conditions differences vary considerably, as well as the average condition at population level varies seasonally and yearly (**Kuriakose, 2017**).

Apart from this research, similar stock assessment done by different authors, using different methods with quite different results. Most of the research makes some assumptions based on commercial sampling and methods, and few scientific campaigns targeted or used the specific methods and equipment adequate to mackerel scad.

3.2.5 Sex ratio comparison

The comparative analysis indicated that sex ratio for the two periods indicated variations of 0.499% and 0.501% for females and males in the first period (Figure 9A), to 0.477% and 0.523% for the second period (Figure 9B), with no statistical significance revealed by the Kruskal-Wallis test (p-value > 0.5). Both sample period were balanced on gender.

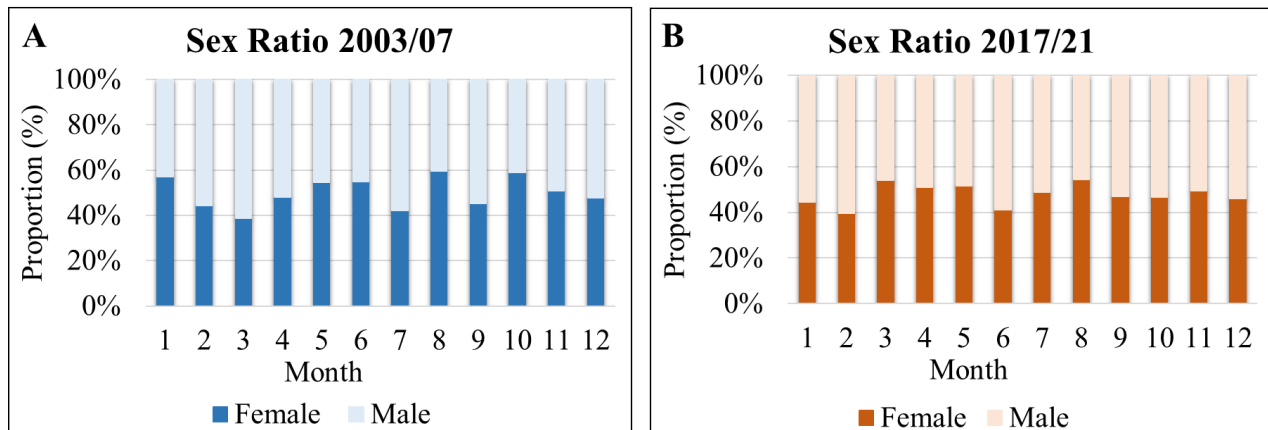


Figure 9: Sex ratio comparison between 2003/07 (A) and 2017/21 (B).

3.2.5 Maturity comparison

The maturity comparison shown that the second period (Figure 10) were sampled more immature than mature mackerel scad, and the differences between the periods have no statistical difference tested by Kruskal-Wallis test (p-value > 0.5). This result can be one indicator that the laws have been respected.

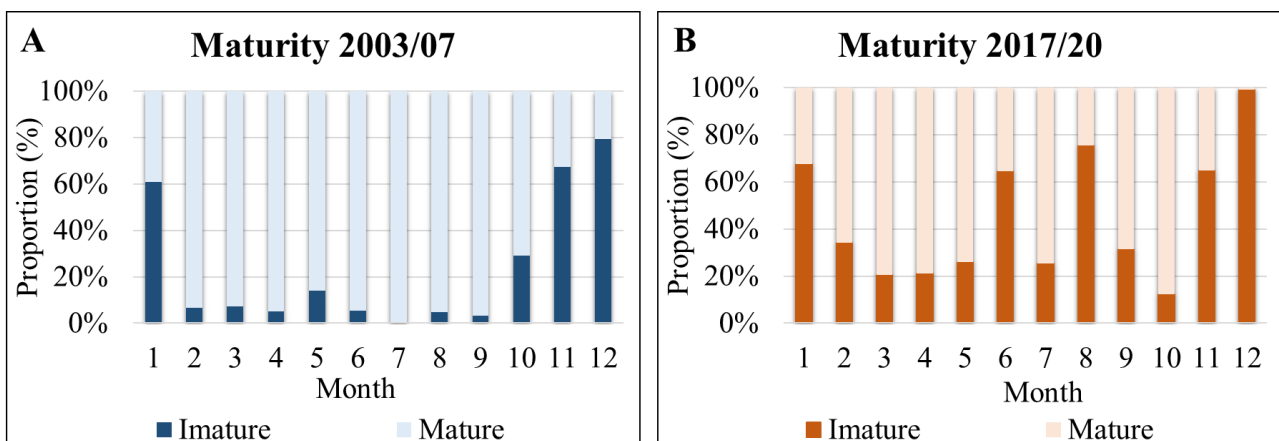


Figure 10: Maturity comparison between 2003/07 (A) and 2017/21 (B).

3.2.4 Gonad maturation state comparison

Comparing the two periods under analysis, it was found that in the first period (Figure 11A), mackerel scad belonging to all levels in the maturity scale used by IMar (annex I) were sampled, but in the second period (Figure 11B), specimens in maturation levels VI and VII were not sampled.

Furthermore, the results indicated that it is possible to find mackerel scad reproducing during the whole year, but with greater representation between February to October, as reported in previous studies by **Jardim (1999)**, **Tariche and Martins (2006)** and **Santos (2018)**. It should be noted that August had just two samples of 75 individuals (from 2020 and 2021), because this period corresponded to BRP of mackerel scad.

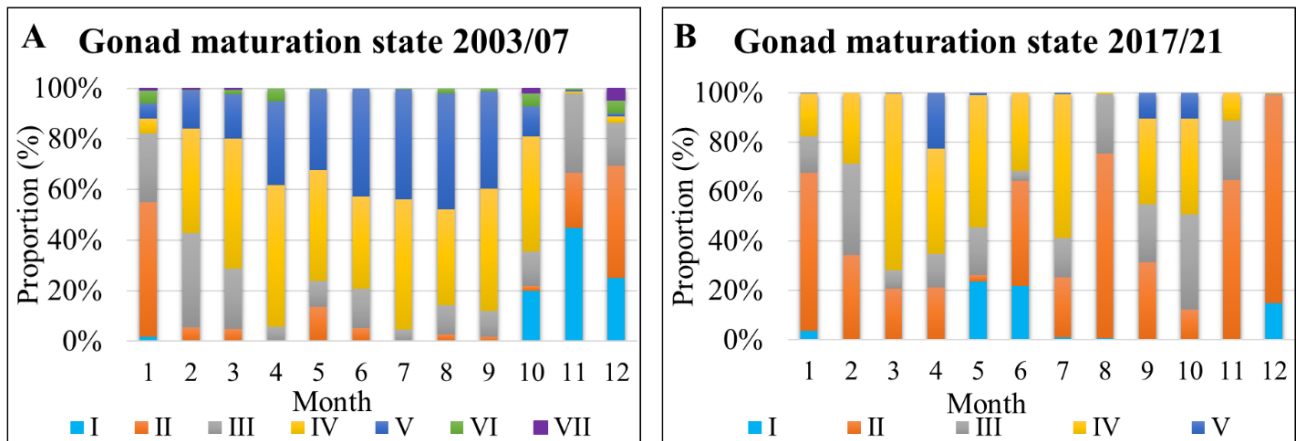


Figure 11: Gonad maturation state comparison between 2003/07 (A) and 2017/21(B).

3.2.5 Gonad somatic index comparison

Mackerel scad in the first period under analysis, had two peaks of gonadosomatic index in the months of April and July, while in the second period, a new peak appeared in the month of October (Figure 12). However, the lack of data on August caused changes on the curve, because only juveniles were sampled. Despite this, the results are in agreement with the results obtained in previous works of **Almada (1997)**, **Santos (2018)** where mackerel scad have two picks of higher maturation and the second is higher than the first. The months that compose the second pick were chosen to implement the biological rest period.

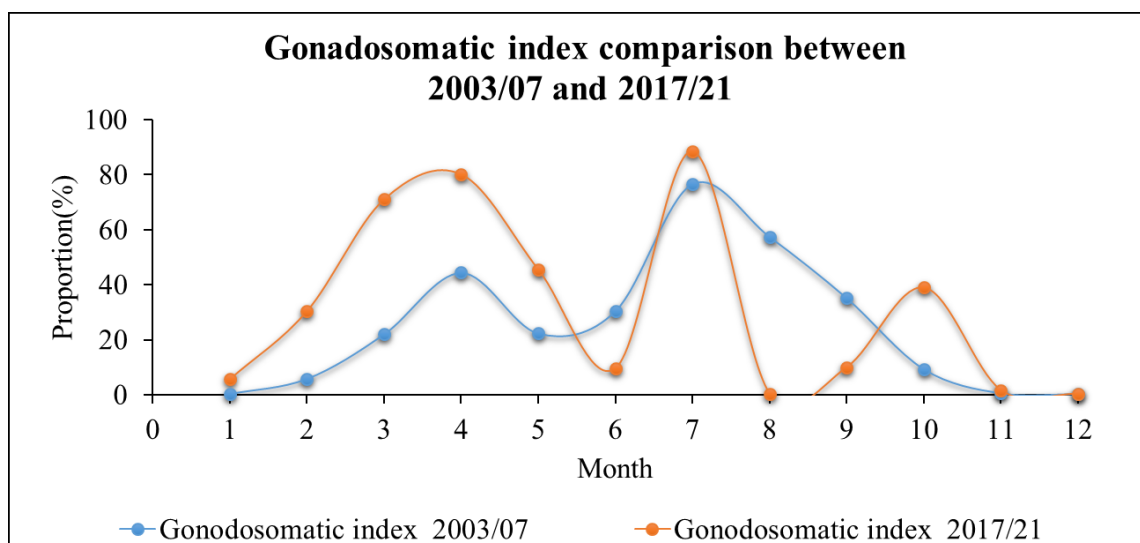


Figure 12: Gonadosomatic Index Comparison between 2003/07 and 2017/21

3.2.6 Maturation ogive comparison

The model results indicate that L_{50} , the first maturity length, increased from 24.2 to 28.2 for the second period in analysis (Figure 13). Smaller values were obtained by **Almada (1997)** $L_{50} = 21.8$ cm FL, **Santos (2018)** $L_{50} = 22.9$ cm FL, **Vieira (2018)** $L_{50} = 20.30$ cm FL, using the same model and data sources in different periods. Histological studies results coming from histological studies of **Costa (2020)** estimated a bigger length of first maturity $L_{50} = 24.1$ cm FL (with samples from 2012 to 2018) similar to the results of the first period in this study.

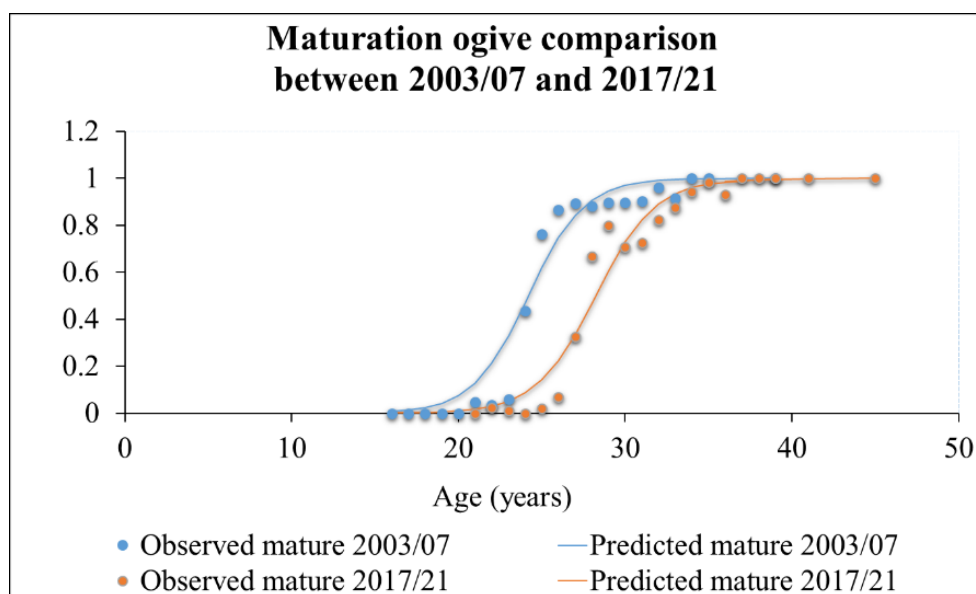


Figure 13: Mature ogive comparison between 2003/07 and 2017/21

Comparing the results, the fishing law resulted on a decrease on mackerel scad fishing pressure, consequently mackerel is reproducing later now or growing more than before. Likewise, as suggested by **Vieira (2018)**, the increase in the sea temperature can be causing these changes on growing and first maturation length of mackerel scad, as **Almada (1997)** reported annual average temperature of 24.1 °C, and **Vieira (2018)** reported 24.7 °C. Relating the L_{50} of mackerel scad with other countries, it was observed that L_{50} varies according to the area and habitat conditions. In Japanese waters, the mackerel scad $L_{50} = 25.8$ (**Shiraishy et al., 2010**) cm LF , and in Hawaiian water, the $L_{50} = 24.5$ cm (**Clarke and Privitera, 1995**).

It is known that climate change affects marine ecosystems by increasing the temperature and acidification (**Cheung et al., 2014**), consequently cause change on ocean physiochemical parameters that can affect primary productivity, species distribution and food web structure (**Cheung et al., 2010**). Each species has his own optimal temperature, habitat and food consumption (**Morgan et al., 2010**), so changes on these optimal conditions impacts directly the fish growth and its biological functions. Likewise, **Chassot et al. (2010)** reported changes on fisheries catch on Large Marine Ecosystems linked to primary production changes on the areas.

3.3 Stakeholders' perception about *D. macarellus* fishing management policies

A total of 175 responses were obtained from the online (n=100) and face-to-face questionnaires (n=75), coming from the 5 focus groups: the fishing institutions, the fishers, the NGO Biosfera I, the academia and the consumers. The analysis showed that gender balance was respected during the questionnaire (Table II). From all groups, 51% of respondents were women and 49% of respondents were men.

Table II: Questionnaire respondent's characterization according to the groups: Gender Balance, Education Level and Years of Knowledge/experience about fisheries.

		Consumer		Fishermen		Fishery institutions		NGO		Academia	
		F	M	F	M	F	M	F	M	F	M
Gender Balance	Gender	63%	37%	51%	49%	43%	57%	31%	69%	67%	33%
Education level	No studies		2%	11%	7%						
	Primary School	6%		31%	30%						
	Professional course		2%		7%		12%				
	High School	8%	5%	5%	9%				19%		
	Bacharel	3%					5%				
	Licence	39%	24%			21%	25%	31%	19%	27%	20%
	Master	11%				10%	17%	10%	19%	3%	20%
	PhD					2%	7%			10%	20%
Knowledge/experience about fisheries	0%	29%	26%				2%		15%	11%	
	01 – 05	21%	5%	3%	2%	9%	2%	23%	23%	39%	11%
	06 – 10	5%		7%	5%	11%	23%	8%	15%		6%
	11 – 15	5%	3%	10%	7%	9%	4%			6%	11%
	16 – 20				5%	6%	2%		15%		
	21 – 25		3%	7%	3%	6%	11%			11%	
	26 – 30	3%		7%	8%	2%	9%				
	36 – 40				7%		2%				
	31 – 35			7%	7%		2%				
	45 – 50			8%	3%						
	50 – 55			2%	2%						

The participants have diverse years of experience/knowledge in fishing. The participant with more years of experience was in the fishers' group with 50-55 years. In general, 36% of people with more years of experience are the fishers, followed by the institutions with 4% of people with 35 to 55 years of experience in fishing while the consumer is the group with the least knowledge/experience about fishing. In contradiction, the fishing institutions and academia have higher levels of education than others (Table II).

Most of the 90% of people that were asked to answer the survey collaborated, from all the target focus groups. Fishers are collaborating more for fishery studies, because they are better instructed and understand the importance of fishing sustainably. According to **Ramos (1999)**, in the

past, it was difficult to involve them in sustainable fishing projects. Therefore, it is clear that people are aware about the importance of this policy.

When asked about their knowledge of the species restoration policies, BRP and MLS, the response showed that most of the respondents are aware of these policies, where the ones most aware of them are the fishery institutions (Figure 14A e 14B). In addition, the policy that they most know was the BRP. It was found that those who are not directly linked as fishing institutions do not know well the BRP and MLS policies, especially the fishers and consumers (Figure 14A e 14B). Similar results were obtained on the SWOT analysis, where academia and NGO groups pointed out that most people do not know the laws very well.

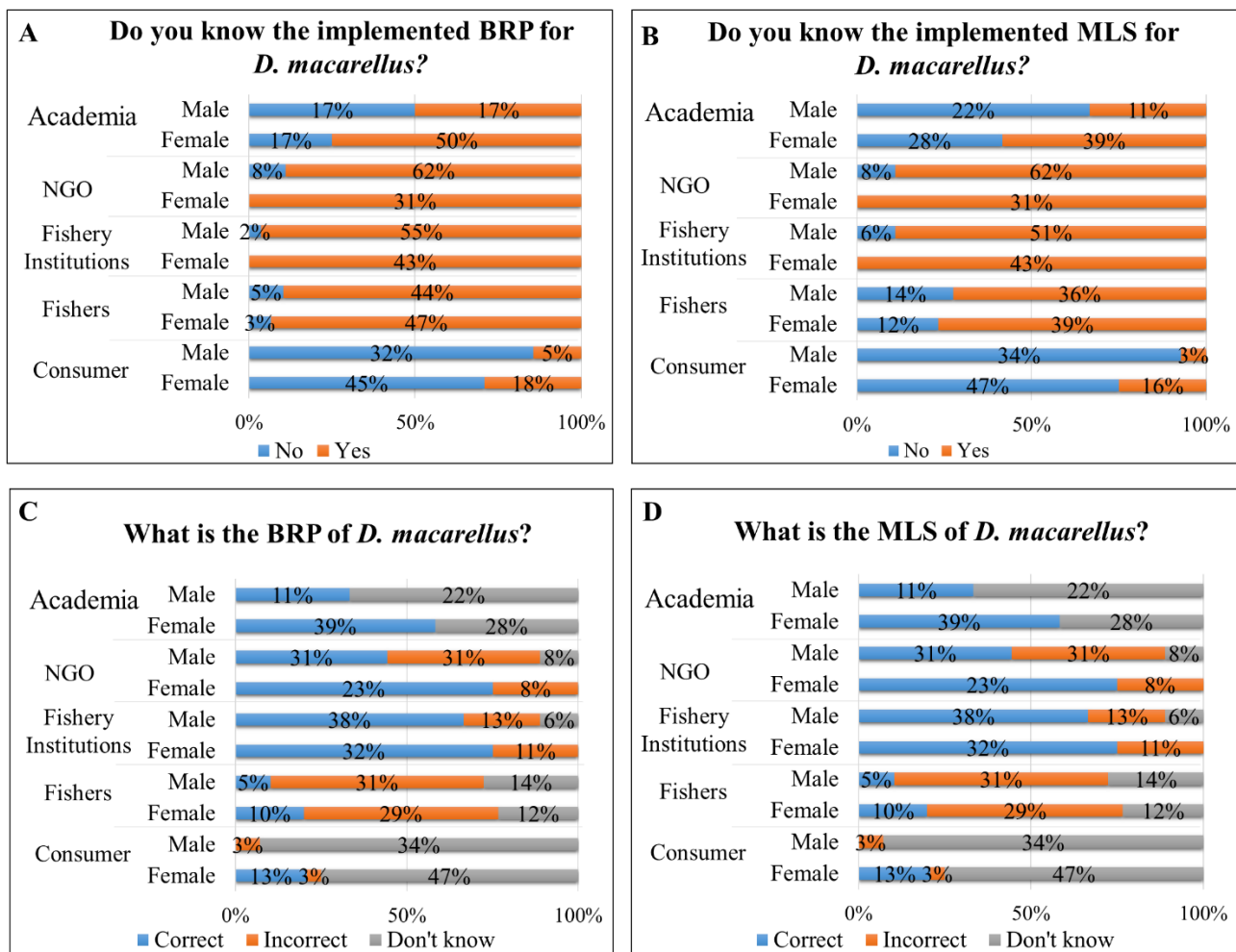


Figure 14: Questionnaire responses percentages about stakeholders' perception and knowledge and about implemented fisheries policies minimum landing size (A) and Biological Rest Periods (B) and responses assessment about Minimum landing size (C) and Biological Rest Periods (D).

Based on fishers' perception of the status of the mackerel scad stock, before and after the implementation of the mentioned fishing policies in 2008, most fishery institutions, NGO and academia responded that the stock was decreasing before the implementation of the policies (Figure 15A). Although fishers claim that before the implementation of the laws, the stock was increasing

because it was easier for them to fish mackerel scad than now while consumers responded that they did not know (Figure 15A). After the implementation of the policies, most of the academia and consumers said to not know the status of the stock (Figure 15B). The fisheries institutions believe that the stock is increasing while the fishers believe that it is decreasing. Landing catch analyses since 1992 to 2020 indicates that mackerel scad landing catch is still decreasing (Figure 9), but on the other hand now mackerel scad have bigger length size (Figure 3).

They were asked to evaluate their knowledge about the policies, and the majority responded that they have a reasonable knowledge of the policies, with the fishing institutions, followed by NGOs and academia saying they know 'well' and 'very well' the policies implemented (Figure 15C). While questioned about their opinion about the laws, most people agreed that they are reasonable or good, and some people from Fishery Institutions, NGO and Academia said that the policies are very good (Figure 15D).

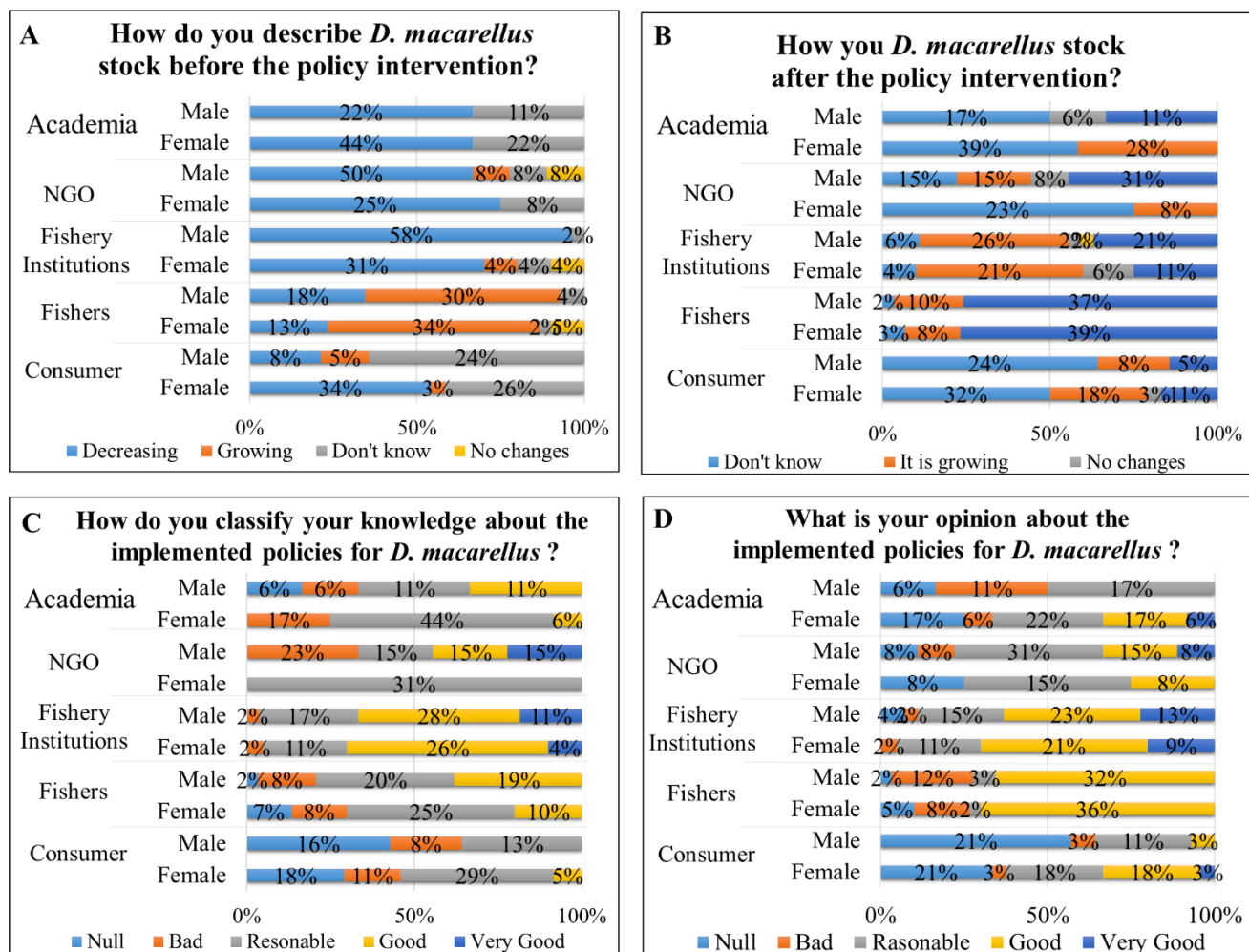


Figure 15: Questionnaire responses percentages about stakeholders' perception and evaluation of implemented fisheries policies

Some people believe that mackerel laws are guiding to increase landing catches (collected on the IMar **Bulletins annual Series from 1992 to 2020**) , but landing catch analyses since 1992 to 2020 indicate that mackerel scad landing catch is still decreasing (Figure 14), but on the other hand, now mackerel scad have bigger length size (Figure 5). At least, the implemented laws have been respected, and mackerel is having time to grow more before being fished. The question now, has asked by the fishers during the workshop was “Where is mackerel scad stock hiding”?, “Why are mackerel scad landing catches decreasing?”, “Are the fishing policies contributing for its disappearance on Landing catches?” “Are Climate Change impacting the catches?”. Many questions were asked, and few answers were provided, but its sure that there is a need to do more studies about its distribution and test possible climate change impacts on mackerel scad stock.

It is clear that most of the people have heard about the laws, but most of them do not know very well about the laws. The ones that know better the laws are the ones working on fisheries institutions. These results are normal, because the fisheries institutions have an obligation to know the laws, while for the others it is not mandatory. Although, there is a need to increase laws dissemination to society, because even some people trust that they implemented the policies to allow mackerel scad to reproduce before being fished, they need to understand the importance and means of the policies for fisheries, food security and consequently for human beings.

While asking them which impacts have they seen on the mackerel scad (Figure 16A), most of the answers were that mackerel scad landing catch decreased while landing size and its economic value increased, as well as, the implemented laws awareness increased. Likewise, were asked respondents opinion about the needed improvement on the mackerel scad policies fishing and measures (Figure 16B), and most of people said that there is a need to improve surveillance and adjust the minimum landing size and the biological rest period



Figure 16: Respondent opinion about laws impact (A) and needed improvements (B) on mackerel scad fishing (word cloud build on R).

Currently, the NGO Biosfera I implemented one project to raise awareness about fishing law policies, targeting some of the fish species that are fished under the implemented fishing laws (**Bisfera 2019**). It helps people to memorise the implemented MLS of the target species using a spot on the back of the public bus (**Biosfera, 2021**). Additionally, **Gonzalez et al. (2020)** consider that for a young state such as Cabo Verde, the legislation gaps on the fishing sector, cause the fisheries management challenges, increased by the lack of monetary and education of fish operators and needs to monitor fishing at all.

It is clear that fisheries management are working to improve caboverdean fisheries, but still missing some gaps and lacks to fill up. Despite all implemented legislations and measures, mackerel scad catch is still decreasing (Figure 17). Some fisheries scientists have linked their disparity to climate change and its consequences. Insular states are more exposed to climate change consequences, due their natural conditions that sometimes cause or make worse its limitations (**Gonzalez et al., 2020**).

Changes in large-scale climate patterns from both anthropogenic and natural causes can affect marine communities (**Geraldi et al., 2019**). Likewise, **Link et al. (2010)** pointed that commercial fishing had a greater impact than climatic indices on soft bottom and pelagic marine ecosystems within the Northern Hemisphere. For example, **Palomares et al. (2021)** estimated *D. macarellus* biomass to be 0.41 B/BMSY in Cabo Verde waters, and the same study shows that mackerels are grossly overfished in Mauritania, Senegal, Gambia, Guinee-Bissau, Guinea, Sierra Leone.

What's more, fishers believe that the cause of the species disparity is the increase of international boats to fish large pelagic fish, also known as predators of mackerel scad in Cabo Verde. According to them, large pelagic fish are responsible for bringing the mackerel scad closer to the coast and in the upper waters where they can fish it. Also, in other cases, due to Illegal and Unreported and Unregulated fishing practiced by international boats, since fishing surveillance is something difficult to do by the Caboverdean authorities due to the lack of acquisition of monetary and other logistical resources. **McNaughton (2008)** proved the same, by showing that large pelagic fishes, such as tuna species can be affecting the ecology and distribution of mackerel scad, as it is predated by tuna species.

Landing catches from 1992 to 2020 from tuna species and mackerel scad were compared. A superficial analysis indicated that the yellow fin tuna landing catches were increasing, all the others catches, inclusive mackerel scad were decreasing (Figure 17). The tuna species landing catch used are coming just from national fishing. If, add the catches from international fishing, 5000 to 8000 tonnes per year, as agreed on the bilateral agreement between Cabo Verde and European Union to

fish tuna species and similar species, including also the shark fishing catches, these values will increase. So, it is possible that large pelagic fishing can be impacting mackerel distribution on the EEZ of Cabo Verde.

Consequently, if overfishing of these mentioned species is happening, the structure of the marine food chain can be affected (**Cheung et al., 2010**). In the absence of predator pressure, the mackerel scad can migrate to other places. In addition, mackerel scad have been found on caches on Canary islands, as well as, other mackerel species, *Decapterus tabl* appeared and still increasing on Cabo Verde landing catches as reported by the National Director of DGP, Dr. Albertino Martins (**Sapo Videos, 2018**).

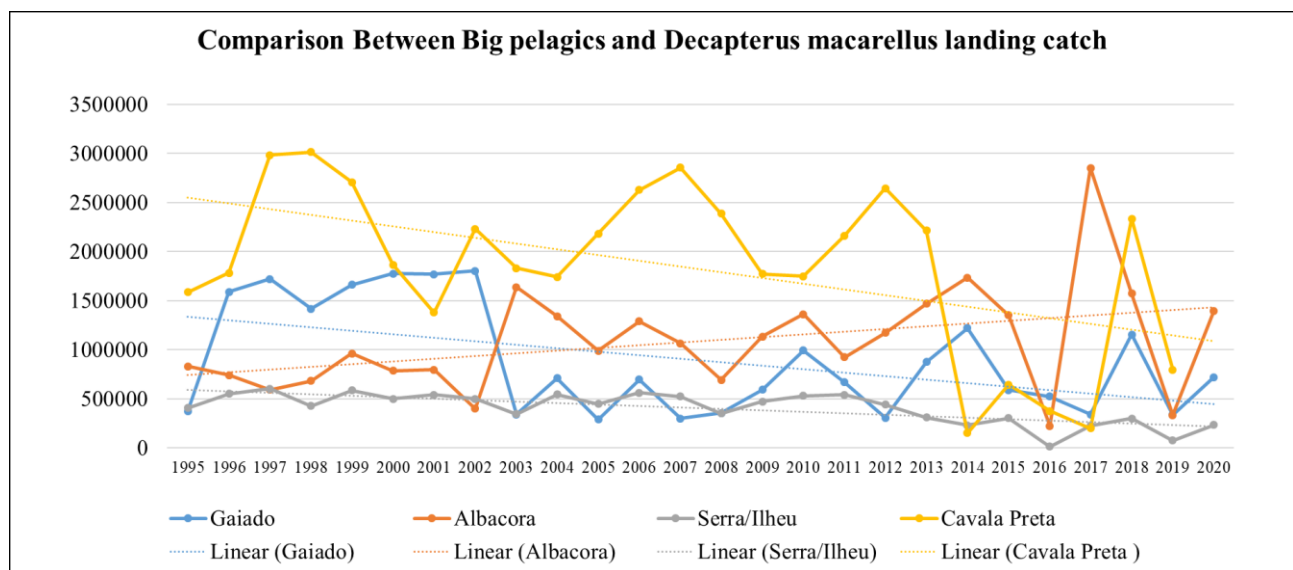


Figure 17: Comparison between large pelagic and *Decapterus macarellus* landing catch data from 1993 to 2020.

Moreover, in other cases, Illegal and Unreported and Unregulated (IUU) fishing practiced by international boats, since fiscalization is something difficult to do by the Caboverdean authorities due to the lack of acquisition of monetary and other logistical resources. Additionally, fishers capacity building projects should be done, encouraging and guiding them to fish others halieutical resources less known and explored such benthic species like scorpionfishes, shrimps, crabs (**Macremar and Mariscomac, 2018**).

Not only legislation and measures should be implemented or revised, but also new and updated studies and assessments should be done to improve fisheries management. For example, coastal zones are vulnerable to anthropic pressures like sand extraction, deposition of sediments, causing changes on these ecosystems (**INDP, 2011b**). These areas are recognised to be potential nursery areas due their topographic and oceanographic conditions (**MAAP, 2004**). Protected nursery areas can lead to

mackerel eggs and juveniles' mortality reduction, consequently increasing recruits biomass to fisheries.

To facilitate this process, fisheries stakeholders need to join forces, as pointed out by **Gonzalez *et al.* (2020)** there has a weak association between universities and fisheries research/management centres. Also, Cabo Verde is one developing country that depends on external help to support its economy. Otherwise, benefits can be taken from the increasing promotion wave of youth entrepreneurs. For example, some projects can be developed, aiming to increase funds to support research while promoting sustainable socioeconomic activities.

CONCLUSION

This research goal was to unravel mackerel scad fishing restoration policies effectiveness on northern Cabo Verde. The study covered the political, ecological and social aspects of these policies. The results showed that Cabo Verde has implemented good policies and measures to support sustainable fishing. Despite the management and resource constraints, Cabo Verde's fishery management is well organised and structured. The government has developed policies based on important international conventions promoting sustainable development. It is visible that they have been doing their best to support policy making and the implementation process.

However, there are some gaps and shortcomings in their management, to improve fishing legislation and effective laws implementation. First of all, to make good policies to better fit marine resources exploitation, there is a need to know very well the ecological resources, distribution, abundance and ecosystem that they are managing. This is the major gap that needs to be filled, because mackerel scad is unknown on most of the biotic and abiotic factors on Cabo Verdean waters context. When it is like that, it will be difficult for fishery management to make adequate and effective laws to suit mackerel scad fishing.

In this research for instance, it became clear that mackerel scad done in different periods revealed that on one hand, its landing catch is increasing while on the other hand the landing abundance is decreasing. Biological rest period effect, is allowing the mackerel scad to escape from fishing and grow more before being fished. Although, it can be that just the bigger mackerel scad has been selected by fishing due the implemented minimum landing size policy, or the policies combination effects, due others unknown biotic or abiotic factors.

Not only anthropogenic factors, like overfishing of large pelagic fish but also climate change impacts on mackerel habitat can be affecting mackerel scad stock on caboverdean waters. It is difficult to assume these assumptions, because there are no studies supporting these hypotheses supported by some fishery stakeholders. Besides, it can be just reflecting mackerel natural behaviour changes on his ecology. Also, other anthropogenic factors can be impacting mackerel life stages like spawning and young stages.

The fishers know about the laws and most of them can explain their importance, because they are the focus of fisheries management, laws enforcement and dissemination campaigns. The fishers' behaviour to the questionnaires showed that most of them are willing to collaborate in the development of fisheries, as they know that they will directly benefit from the implementation of the right measures. It was also found that many fishers are aware of the laws, but do not know the essence of these measures to recover fish stocks. Despite of this, good number of fishers just see that these

policies are just causing reduction on their income, or even some believe that they are the cause of mackerel scad disappearance on fishing grounds

The study concludes that all fishery stakeholders are aware about the fisheries policies, because most people have heard it, but not all of them can understand and explain why the implementation of these policies and how important they are for sustainable fishing. For example, even having some gaps on laws dissemination, consumer group, the most disconnected from fishing laws, also know about the laws and some know about their importance for fisheries because they are linked to one of the other groups, but they cannot explain why the implementation of these policies.

The existent difficulties in financing scientific studies in Cabo Verde, are the major cause of scientific knowledge lack and gaps. Many projects are pointed as needing to be done, but this is impossible without funding.

RECOMMENDATIONS

This research goal was to unravel mackerel scad fishing restoration policies effectiveness. The findings showed that Cabo Verde fisheries management is very well structured, and its legislation is based on sustainable resource exploitation and the fisheries institutions are designed to support its implementation. Besides, important results have been observed, on one hand mackerel scad is increasing in size but, on the other hand mackerel scad landing catch is decreasing. That's why at the end of this research we would like to propose some recommendations to fishery stakeholders:

1. Fishery Institutions

The results showed that there is a need to know better mackerel scad, to be able discover what is really happening with its stock, as it was one of the most captured commercial fish in the past, but unfortunately, in the known days its landing catches decreased drastically.

Moreover, first, there is a need to better know mackerel scad ecology and distribution to better adjust these policies to mackerel scad fishing. For that, more fish campaigns should be done focused on study the ecology and distribution of mackerel in Cabo Verde, since the nursery areas, larvae dynamics and migration process, recruitment and reproduction areas. Also, those areas need to be evaluated to see if they are being impacted by anthropogenic actions and further responses of climate change. In addition, the cause of mackerel scad disappearance should be analysed, to see if changes in abiotic factors are leading to mackerel scad migration to new fishing grounds and habitats.

In addition, IMar commercial sampling should be updated to minimize commercial sampling influence on the results. For example, if a model can be developed to assess the possibilities of combining mackerel fishing effort, (taking into account all fishing effort), persistence of fishing on the same fishing ground, the mackerel abiotic preferences. This will be easier to predict mackerel ecological behaviour, consequently measures will fit better its exploitation.

For policies to be effective, they should be known. There is a need to improve law enforcement and dissemination by the fishery institutions, because it's clear that fishers and consumers also know about the laws and some aspects about their importance for fisheries. Even though, there is a need to do more policies dissemination actions to help the fishers to understand that even if they face some income reduction during the biological rest period, the revenue in the future will be that they will have mackerel to fish during many years, despite of fish them all in few years and cause mackerel stock decline.

Therefore, to ensure the success of all efforts and policies, laws enforcement should be revised and improved. So far, the results show that the laws have been respected by fishermen, but still some needed adjustments to do, like expand the surveillance to the other landing ports and sites. So, we

should take advantage of the existing national and international partnership with NGO's, universities, investors, entrepreneurs and other relevant partners to fund and design research solutions that fit the country's economy, while generating income and job positions, at the same time contributing to unemployment reduction and marine research development.

2. Fishers

The majority of fishers collaborate in the policy formulation and they respect laws. But, there is a need for more collaborative actions, like capacity building, more cooperation between them and other stakeholders and more workshops/seminars in order to keep them at the same level of knowledge regarding fisheries state, because they are the direct stakeholders and most important author ensuring the sustainable fish exploitation.

Fishers should be more open and active to these actions, by collaborating more in such activities, because even if the government cannot give them subsidies to help them during the biological rest periods, there are many capacity-building activities to help them to find other solutions in these periods.

Fishers complained that with the available fishing materials, it is difficult to catch mackerel scad, so there is a need to improve their fishing material controlled by the fishery institutions.

There is a need to do more policy dissemination actions to help them to understand that even if they face some income reduction due the biological rest period, the revenue will be that the grandsons of their grandsons will have mackerel scad to fish.

3. Academia

Academia plays an important role for research in Cabo Verde, as the government has many limitations in funding the necessary studies to support the policy-making process. UNICV and UTA universities provide very good graduation opportunities focusing on marine research, but the lack of monetary funds makes it difficult to implement research projects. Worldwide, we can find a lot of funding opportunities that can help develop research in Cabo Verde. So, the Academia can motivate and help their students to look and apply for such opportunities focusing on fisheries research in Cabo Verde. This can be helpful in various ways, by contributing for research development, capacity building and at the end of the day, to policy improvement.

4. NGO

The NGO, Biosfera I, is doing a great job, by promoting the activities of law dissemination. Additionally, they developed a cooperation with the villages, to encourage fishers to respect the laws and fish sustainably (Biosfera, 2020). If more projects like these can be implemented by NGOs, to

alleviate state responsibilities, civil society awareness will be raised, and food security for future citizens can be guaranteed, and others SDGs can be achieved on Cabo Verde

PERPECTIVES

This research tried to unravel mackerel scad fishing restoration policies effectiveness on northern Cabo Verde. It covered the political, ecological and social aspects of these laws. It is clear that some efforts have been made by all fishery stakeholders' to support the policy-making and implementation process. Unfortunately, master research cannot cover all the aspects behind the topic, to answer the entire question that appeared during 6 months of research. In this case, we made some perspectives that can improve this work in the future, in order to give more inputs on this topic.

First of all, as Cousteau use to say, “we protect what we understand”, so to make the implemented fishing policies more effective, we need to do more ecological studies to understand better mackerel scad, consequently implement more effective policies for its stock restoration, such as:

- To do more studies regarding its biology and better now its ecology on caboverdean waters.
- Study mackerel scad distribution on Cabo Verdean waters;
- Map the fish nursery areas around the islands and try to understand the anthropogenic changes are influencing fish spawning and larval/ juveniles' life stages;
- Study how climate changes is changing the abiotic parameters on the sea and how this changes are impacting on mackerel scad ecology and distribution;
- Study the prey-predator relationship between the mackerel scad and large pelagic, to better understand the impact that large pelagic fishing has on mackerel scad distribution and migration, during its life.
- In addition, the stock assessment methodologies need to be adjusted to the specific case, by integrating more abiotic and biotic factors in the methodology, in order to be able to provide as much as possible, more realistic results to support policy reformulation.
- Expand this research by including more samples, coming from other islands. Also, replicate this study to other commercial fishes, because if mackerel scad landing catch is decreasing, this can lead to increasing fishing pressure of other exploited fishes, especially the demersal fish that are more vulnerable to fishing.

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APPENDIX

Appendix I: Fish sampling sheet

Biological Sampling Formulary

Specie: _____ Fishing site: _____

Sampling date: _____ Total Sampling: _____

Sampler: _____ Register : _____

N°	Furcal Length (cm)	Weight(g)	sex	Gonad Maturation	Gonad Weight	Observations
1						
2						
3						
4						
5						
6						
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Appendix II: Survey for fishery stakeholders

Fishery stockholders perception evaluation about mackerel scad fishing policies

A Master student at the Science-Policy Interface of Biodiversity and Ecosystem Services for West African Development conducts this survey. One of the objectives of the survey has a social aspect that is related to this topic 'Assessment of the perception of black mackerel scad fishing policies' and will be applied to the general population.

The data that will be provided is absolutely confidential and anonymous and will be used exclusively for academic and scientific purposes. We ask you to cooperate by completing the questionnaire individually and to be as accurate as possible in its completion.

Thank you in advance for your collaboration.

A. Please, fill the spaces with the requested information about you:

Age ____ years	Gender: M ____ F ____	Academic level _____	Years of experience _____	Occupation _____
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B. Please, provide the following answers according to your knowledge about the policies implemented for the restoration of fish stocks in Cabo Verde :

1. Have you heard about the implemented policies for fish stock restoration in Cabo Verde?	Yes____ No____
2. Do you know the implemented policies for fish stock restoration in Cabo Verde?	Yes____ No____
3. How would you rate your knowledge about the implemented policies for fish stock restoration?	Null____ Bad____ Razonable____ Good____ Very good____

C. Please, provide the following answers according to your knowledge about the policies implemented for the restoration of mackerel scad stocks in Cabo Verde.

1. Do you know the implemented fishing laws for mackerel scad in Cabo Verde?	Yes____ No____
1. If yes, what are they? R:_____	
2. What is your opinion about the implemented fishing laws for mackerel scad in Cabo Verde?	____Null ____Bad ____Reasonable ____Good ____Very good
3. Do you know the minimum landing size of mackerel scad?	____Yes ____No
4. If yes, what is it? R:_____	
5. In case fishers fish mackerel scad smaller than the allowed fishing size, what do they do with them? ____ They are confiscated; ____ They are returned to the sea; ____ They go to fisheries research; ____ They are donated to social institutions; ____ They go to the fish market and are sold afterwards; ____ Other : _____	
6. Are the fishing laws well implemented and respected?	Yes____No____ Don't know____
7.1. If they do not respect the laws, are they punished?	Yes____No____ Do no know____
6.2 . If yes, what are these punishments? ____Tax ____Jail ____Vessel confiscated ____Other	
7. Do you know the mackerel scad biological rest period?	Yes____ No____
8.1. If yes, when exactly? R:_____	
8. The biological rest period law is respected?	Yes____No____ Don't know____
9.1. If they do not respect the laws, are they punished?	Yes____No____ Don't know____
9.2. If yes, what are these punishments? ____Tax ____Jail ____Vessel apprehension ____Other	
9. How do you describe the mackerel scad catch before laws implementation?	____Decreasing; ____Increasing ; ____No changes ____Don't know;
10. How do you describe the mackerel scad catch after laws implementation?	____ Still decreasing; ____ Still increasing ; ____No changes ____Don't know;

10.1. If it is decreasing, which measures can you suggest?	
R: _____	

10. Have you seen others impacts on mackerel scad catch?	Yes____No____ Don't know _____
11. If yes, which ones?	
R: _____	

Appendix III: Stakeholders workshop

ASSESSMENT OF MACKEREL SCAD FISHING POLICIES:

- Minimum Landing Size and Biological Rest

This workshop is being held with the objective of reviewing fisheries management policies for restoring the *Decapterus macarellus* (black mackerel) stock, closure and minimum catch size. The target audience will be representative members of all fisheries stakeholder groups: fishers, fisheries research, management and inspection institutions, the university and school of the sea, and NGOs dealing with this issue.

The method to be used is SWOT (Strength, Weakness, Opportunities, Treats) analysis, i.e. the strengths, weaknesses, opportunities and threats that can be identified in the implemented black mackerel fishery laws, more precisely the minimum catch size and closed season. Participants will be organized into 4 groups according to their role in the fishery, and each group will be asked to apply the analysis, based on their knowledge of the subject. Each group will be assisted and guided in the execution of the work.

After the group work, each group will present the results according to their point of view, after will be discussed with the other groups.

The seminar will have a maximum duration of 2 hours, and will follow the following program:

TIME	ACTIVITIES	
14:15	Invites reception	IMar Department of Halieutical Research and Biodiversity Director (Dr. Vito Ramos)
14:30	Session opening	Master student
14:40	Master Research Project presentation	Master Student
14:50	Group work: SWOT analysis	All participants
15:20	Plenary session	All participants
16:20	Closing remarks	Master Student
16:30	Coffebreak	
16:45	Closing	

Appendix IV: Stakeholders workshop presence list

SEMINARIO

AVALIAÇÃO DAS POLÍTICAS DE PESCA DA CAVALA PRETA:

- Tamanho mínimo de captura e época de defeso

LISTA DE PRESENÇA

	NOME	INSTITUIÇÃO	FUNÇÃO	ASSINATURA
1	Orvelino Duarte	UTA	Do Ceu P Pro. Belos	SL
2	João Augusto Duarte	Oca I	Armador	João Augusto
3	Alfonso Duarte	A Pesc	Armador	Alfonso
4	Alfonso Duarte	A Pesc	Armador	Alfonso
5	Alfonso Duarte	A Pesc	Armador	Alfonso
6	Alfonso Duarte	A Pesc	Armador	Alfonso
7	Alfonso Duarte	A Pesc	Armador	Alfonso
8	Alfonso Duarte	A Pesc	Armador	Alfonso
9	Alfonso Duarte	A Pesc	Armador	Alfonso
10	Alfonso Duarte	A Pesc	Armador	Alfonso
11	Alfonso Duarte	A Pesc	Armador	Alfonso
12	Alfonso Duarte	A Pesc	Armador	Alfonso
13	Alfonso Duarte	A Pesc	Armador	Alfonso

14	Jessica de Mato	Biosfera	Assistente de Projeto	Jessica de Mato
15	Delva Fortes			Delva Fortes
16	Christiane S.S. Alves	Imar	Biologa Estagaria	Christiane
17	Nara Morais	UTA		
18	Fulvia Duarte	UTA		Fulvia
19	Laura da Luz	UTAR		Laura
20	Katylene da Cruz Belgado	UTHB	IMAR Interm	Katylene Belgado
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ANNEX

Annex I: Maturation Scale Guide for Pelagic Fish (Translated from IMar scale)

Maturation state	Female	Male
Immature (1)	Indifirenciated	Indifirenciated
Sexual rest (2)	Small, transparent or light pink gonads, oocytes invisibles.	White or slightly translucent, very fine, knife-shaped gonads
In Maturation (3)	Gonads vary in colour from pale pink to light orange, some oocytes are perfectly visible through the ovarian membrane	White gonads, no liquid emission after incision
Pre-spawning (4)	Thicker gonads. The oocytes are visible through the ovarian membrane making the ovarian surface granular	Gonads softer and whiter, liquid emission after incision
Spawning (5)	Gonads very thick, ovarian membrane very thin. The oocytes are hyaline and sticky	Thick and soft gonads, sperm emission in case of the fish abdomen is pressed.
Post-spawing (6)	Flaccid ovary and very vascularised, with numerous hyaline spaces	Flaccid testicles with fine vascularisation
Gonadal regression	Very flaccid ovary and very vascularised,	Very flaccid testicles very vascularised

Summary

Fisheries management in Cabo Verde, since young years after independence and republic constitution, tried to implement fisheries laws to support marine resources sustainable exploitation, preventing overfishing. Unfortunately, until now, there was no opportunity to evaluate these policies' effectiveness. Therefore, the aim of this research is to evaluate the effectiveness of the implemented policies regulating the *Decapterus macarellus* (mackerel scad) fishery by analysing these fisheries policies true literature review and SWOT analysis, assessing mackerel scad stock before and after the implementation of the policies and evaluating the perception of fisheries stakeholders on these policies. The results showed that one hand mackerel scad landing size has increased and on the other hand the landing catches are still decreasing. Moreover, the stakeholders know the policies and they have also brought positive contributions to fisheries. Despite this, there are some gaps in the policies and lack of research on ecology and distribution of mackerel scad. Furthermore, there is a need to understand the causes of the decline in catches, to have a better understanding of the mackerel scad fishery. Ultimately, all this research will support policy reformulation for better implementation of sustainable measures to improve the ecosystem services provided by its fisheries.

Keys words: Science-Police; mackerel scad; fisheries policies;

Résumé

La gestion de la pêche au Cabo Verde, depuis les premières années après l'indépendance et la constitution de la république, a essayé de mettre en œuvre des lois sur la pêche pour soutenir l'exploitation durable des ressources marines, en empêchant la surpêche. Malheureusement, jusqu'à présent, il n'y a pas eu d'opportunité d'évaluer l'efficacité de ces politiques. Par conséquent, l'objectif de cette recherche est d'évaluer l'efficacité des politiques mises en œuvre pour réglementer la pêche de *Decapterus macarellus* (chinchard) en analysant ces politiques de pêche par une véritable revue de la littérature et une analyse SWOT, en évaluant le stock de chinchard avant et après la mise en œuvre des politiques et en évaluant la perception des acteurs de la pêche sur ces politiques. Les résultats ont montré que, d'une part, la taille des débarquements de maquereaux a augmenté et que, d'autre part, les captures au débarquement continuent de diminuer. En outre, les parties prenantes connaissent les politiques et elles ont également apporté des contributions positives à la pêche. Malgré cela, il y a des lacunes dans les politiques et un manque de recherche sur l'écologie et la distribution du chinchard. En outre, il est nécessaire de comprendre les causes du déclin des captures, afin de mieux comprendre la pêche du chinchard. En fin de compte, toute cette recherche soutiendra la reformulation des politiques pour une meilleure mise en œuvre de mesures durables afin d'améliorer les services écosystémiques fournis par ses pêcheries.

Mots clés : Science-Police ; chinchard ; politiques de pêche ;